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Some Topics in Ergodic Ramsey Theory

Master Thesis

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Abstract

This paper is a survey which focuses on a number of concepts in ergodic Ramsey theory, including but not limited to sets of recurrence, sets of strong recurrence and van der Corput sets. In particular, we review the connections between these families of sets. For instance, we show that van der Corput sets are sets of recurrence. We also review Kamae and Mendès France's sufficient conditions for being a van der Corput set, Ruzsa's proof that the aforementioned conditions are also necessary, and Mountakis' proof that there is a set of strong recurrence that is not van der Corput. In the last section of the paper, we provide an exposition of two proofs of Roth's theorem on arithmetic progressions, the second of which is ergodic in nature and discusses a general version of Roth's theorem for countable abelian groups.

*Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth;*

*Then took the other, as just as fair,
And having perhaps the better claim,
Because it was grassy and wanted wear;
Though as for that the passing there
Had worn them really about the same,*

*And both that morning equally lay
In leaves no step had trodden black.
Oh, I kept the first for another day!
Yet knowing how way leads on to way
I doubted if I should ever come back.*

*I shall be telling this with a sigh
Somewhere ages and ages hence:
Two roads diverged in a wood, and I—
I took the one less traveled by,
And that has made all the difference.*

- *The Road Not Taken, Robert Frost.*

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I wrote this thesis in perhaps the most difficult semester of my education thus far, where I questioned where I should continue pursuing mathematics or give up entirely. I still don't know what the future holds, but being able to make it to the finish line would have been completely impossible without the support of my friends and parents, who kept my spirits up and allowed me the luxury of working without having to worry about many other things in life. Thanks to the Zurich boys - Raphi, Florian, Sam, Adam, Ulysse, Titouan - for letting me visit any time, to the Geneva boys - Mohammed, Boris, Najeeb, Emre, Ivan, Diego, Jack, Dan - for all the laughs and love, and to my family for looking after me every day. It means more than I can put into words.

Maybe the single most important factor in ensuring I didn't fall out of love with mathematics entirely was the Swiss Mathematical Olympiad, which reminds me every day - through my own efforts and the achievements of my students - why I pursued math in the first place, and gives me a reason to not give up yet. Thank you to all my collaborators - particularly Valentin, Arnaud, David, and Marco - for all you do to make the SMO tick, and to all the kids for making us prouder each year with your hard work, and inevitably, your growing success.

Usually, one finishes a major project like this with a sense of triumph, and perhaps pride. I finish it with a strange sense of sadness, because for the first time in my life, I will not begin the new academic year as a student. I hope that during my year of civil service I will find a way forward to continue doing what I love, and if I have learnt one thing about myself in the past few months, it's that I'm not going to give up on it yet. After all, the journey would be quite boring without some turbulence, and I do have a reputation to uphold as the endlessly overconfident optimist. A storm might be coming, but I'll see you on the other side!

Contents

Contents	iv
1 Introduction	1
2 Preliminaries	4
2.1 Notions of largeness for sets of integers	4
2.2 Measure-preserving systems and recurrence	9
2.3 Harmonic analysis	10
2.4 Equidistribution and the van der Corput theorem	12
3 Sets of recurrence and van der Corput sets	15
3.1 Sets of recurrence	15
3.1.1 Intersectivity	16
3.1.2 Partition regularity	20
3.2 Van der Corput sets	21
3.2.1 Van der Corput's difference theorem	21
3.2.2 Van der Corput sets are sets of recurrence	22
3.2.3 Ruzsa, Kamae and Mendès France's characterization of van der Corput sets.	24
3.3 Sets of recurrence that are not van der Corput	34
3.3.1 Generalisations and more Fourier analysis	34
3.3.2 Mountakis' result	39
4 Roth's theorem on arithmetic progressions in different contexts	54
4.1 A combinatorial proof of Roth's theorem	54
4.1.1 The equivalence of finitary and infinitary Roth	54
4.1.2 Szemerédi's proof of Roth's theorem	56
4.2 Bergelson's proof of Roth's theorem for abelian countable groups	60
4.2.1 Preliminaries	60
4.2.2 Bergelson's proof	64

Bibliography	68
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Chapter 1

Introduction

Ergodic theory is a major branch of mathematics that seeks to understand and analyse properties of dynamical systems that describe the behaviour of points in ambient space, usually in a context relating to mathematical physics. For instance, fluid dynamics, Brownian motion and the behaviour of pendulums can all be described via a dynamical system of some kind. Of particular interest is the long-term behaviour of such systems. Consider for example the following result, first appearing in a fundamental paper by Poincaré[1] in 1890, which, in modern terms, can be formulated as follows.

Theorem 1.1 (Poincaré recurrence theorem) *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving dynamical system. For any $A \in \mathcal{B}$ such that $\mu(A) > 0$, the set of points $x \in A$ such that there exists $N \in \mathbb{N}$ with $T^{-n}(x) \notin A$ for all $n > N$ has measure 0.*

A note to readers unfamiliar with the terms in this introductory chapter - they will be formally defined in Chapter 2.

In other words, this theorem states that for almost every point $x \in X$, the sequence $x, T^{-1}(x), T^{-2}(x), \dots$ will "visit" A infinitely many times. What if we now consider the sequence $A, T^{-1}A, T^{-2}A, \dots$ instead? Does it also "return" to A ?

Definition 1.2 (Set of returns) *Given a measure-preserving dynamical system $(X, \mathcal{B}, \mu, T)$ and a set $A \in \mathcal{B}$ such that $\mu(A) > 0$, we define the set of returns as*

$$R_A := \{n \in \mathbb{N} : \mu(T^{-n}A \cap A) > 0\}.$$

This definition leads to many natural questions: what properties does the set of return times have - does it have positive density? Is the gap between return times necessarily bounded? Does it contain multiples of every number?

Another version of the Poincaré Recurrence Theorem states that for any set A with positive measure, the set of return times R_A is nonempty. This version

of the Poincaré Recurrence Theorem leads to the following natural definition, due to Furstenberg[2].

Definition 1.3 (Set of recurrence) We call $R \subset \mathbb{N}$ a set of recurrence if it has the following property: given a measure-preserving dynamical system $(X, \mathcal{B}, \mu, T)$ and a set $A \in \mathcal{B}$ such that $\mu(A) > 0$,

$$R_A \cap R \neq \emptyset.$$

Let us take a brief trip to another domain of mathematics - that of additive combinatorics. This field studies subsets of the integers and their properties under arithmetic operations. Additive combinatorics is closely related to Ramsey theory, a branch of combinatorics that espouses the philosophy that if one takes a sufficiently large structure and partitions it into disjoint pieces, one of the pieces always displays certain properties. One celebrated result in the intersection of additive combinatorics and Ramsey theory is van der Waerden's theorem, which states that if one colours the elements of $\mathbb{N}$ in a finite number of colours, one can find arbitrarily long monochromatic arithmetic progressions.

Roth[3] proved that any subset of $\mathbb{N}$ with positive upper density has a 3-term arithmetic progression. Szemerédi[4] established a vast improvement on Roth's theorem by showing that subsets of $\mathbb{N}$ with positive upper density have arbitrarily long arithmetic progressions, a result which implies van der Waerden's theorem. Furstenberg and Katznelson[5] in turn established a multidimensional generalization of Szemerédi's theorem using ergodic theory. Furstenberg, with his collaborators, proved many additional results in Ramsey theory using ergodic methods. Naturally, this field of study is today called ergodic Ramsey theory, and some of these results still do not have conventional combinatorial proofs. The principle pioneered by Furstenberg which allows the derivation of combinatorial statements from ergodic results is today called Furstenberg's correspondence principle (see Bergelson[6]).

Theorem 1.4 (Furstenberg's correspondence principle) Let $E \subseteq \mathbb{N}$. There exists a measure preserving system $(X, \mathcal{B}, \mu, T)$ and a set $S \in \mathcal{B}$ with $\mu(S) = \bar{d}(E)$ such that for any $n_1, \dots, n_k \in \mathbb{N}$, we have

$$\mu(A \cap T^{-n_1} A \cap \dots \cap T^{-n_k} A) \leq \bar{d}(E \cap (E - n_1) \cap \dots \cap (E - n_k)).$$

Finally, our metaphorical introductory journey arrives at a result due to van der Corput[7], on equidistributed sequences.

Theorem 1.5 (van der Corput difference theorem) Let $(x_n)_{n=1}^{\infty}$ be a sequence of reals. If for all $h \in \mathbb{N}$, the sequence $(x_{n+h} - x_n)_{n=1}^{\infty}$ is equidistributed modulo 1, then so is x_n .

An informal definition of equidistribution is that for any interval I of $[0, 1]$, the proportion of the sequence taking values in I should be equal to the length of I - so half the values are in $[\frac{1}{4}, \frac{3}{4}]$, for instance.

One might wonder whether we can weaken the condition in van der Corput's difference theorem to all h in some strict subset H of $\mathbb{N}$, rather than needing it to hold for all positive integers. Indeed, Kamae and Mendès-France proved that such subsets existed and named them van der Corput sets[8], which we will discuss in great detail in this paper.

At first glance, there is no apparent connection between sets of recurrence and van der Corput sets, but Kamae and Mendès-France[8] showed that for any set of returns R_A and any van der Corput set H , $R_A \cap H \neq \emptyset$, implying that all van der Corput sets are sets of recurrence. It is natural to ask whether all sets of recurrence are van der Corput sets and the answer is in the negative, as was shown by Bourgain[9]. Mountakis[10] adapted Bourgain's methodology to answer a question of Bergelson and Lesigne[11] whether there is a set of strong recurrence that is not van der Corput.

This paper will provide an exposition of some results in ergodic Ramsey theory with the necessary preliminaries. The reader is expected to be familiar with the basics of measure theory, functional analysis and harmonic analysis, but no prior knowledge of ergodic theory is expected and all relevant results will be stated and explained. A quick roadmap: in Chapter 2 most of the necessary prerequisite definitions are introduced and pertinent general results are formulated; Chapter 3 is an exposition of Kamae, Mendès-France, Ruzsa and Mountakis' results, and Chapter 4 provides two proofs of Roth's theorem.

Chapter 2

Preliminaries

This chapter establishes some mathematical conventions that we follow and contains most of the basic definitions that are needed going forward.

2.1 Notions of largeness for sets of integers

To begin with, let us lay down some conventions. The set of natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$ does not include 0. If we ever need to refer to the set $\{0, 1, 2, 3, \dots\}$ we will denote it by $\mathbb{N}_0$. We will also use the standard definition $[N] = \{1, 2, 3, \dots, N\}$. Given two subsets $S, T \subseteq \mathbb{N}$, we have the notation

$$S - T := \{s - t : s \in S, t \in T\} \cap \mathbb{N}$$

and similarly, if $n \in \mathbb{N}$ then $S - n$ is an abuse of notation for $S - \{n\}$. The torus $\mathbf{T}$ refers to the quotient group $\mathbb{R}/\mathbb{Z}$.

Defining some notion of size for an infinite subset of $\mathbb{N}$ in a natural way is a central tenet of additive combinatorics. For instance, it is very appealing to say that the even numbers constitute half of the integers, or that the square numbers make up almost none of the integers, but we need a precise way to express this proportionality.

The definition of *density* is probably the first one that comes to mind: simply look at the finite proportions and then take the limit.

Definition 2.1 (Density) *The density of a set $A \subseteq \mathbb{N}$ is given by*

$$d(A) := \lim_{N \rightarrow \infty} \frac{|A \cap [N]|}{N}$$

when this limit exists.

Note that the density does not necessarily exist - for instance, if A is the set of numbers whose first decimal digit is 1, it is easy to see that the density of

the set in question oscillates between $\frac{1}{9}$ and $\frac{5}{9}$. To extend the definition to all sets, we often use *upper density*.

Definition 2.2 (Upper and lower density) *The upper density of $A \subseteq \mathbb{N}$ is given by*

$$\bar{d}(A) := \limsup_{N \rightarrow \infty} \frac{|A \cap [N]|}{N}.$$

The lower density $\underline{d}(A)$ is defined analogously with the $\liminf$.

Note that it is easy to construct sets with arbitrary upper and lower densities. The density, when it exists, is additive - if $A, B \subseteq \mathbb{N}$ and $A \cap B = \emptyset$, then $d(A \cup B) = d(A) + d(B)$, but this is not true for the upper or lower density. The upper and lower densities, however, are subadditive ($A \cap B = \emptyset \implies \bar{d}(A \cup B) \leq \bar{d}(A) + \bar{d}(B)$) and superadditive ($A \cap B = \emptyset \implies \underline{d}(A \cup B) \geq \underline{d}(A) + \underline{d}(B)$) respectively.

Another natural property of a set $A \subseteq \mathbb{N}$ that expresses a notion of size is *syndeticity*.

Definition 2.3 (Syndetic set) *A set $A \subseteq \mathbb{N}$ is syndetic if there exists $n \in \mathbb{N}$ such that, for any $N \in \mathbb{N}$ we have*

$$A \cap \{N, N+1, \dots, N+n-1\} \neq \emptyset.$$

One well-known example of a set that is not syndetic is the prime numbers: for any $n \in \mathbb{N}$ the numbers $(n+1)! + 2, (n+1)! + 3, \dots, (n+1)! + n + 1$ are divisible by $2, 3, \dots, n + 1$ respectively.

Lemma 2.4 *Syndetic sets contain arbitrarily long arithmetic progressions.*

We will prove this lemma by taking for granted one of the first major results in Ramsey theory - van der Waerden's theorem[12].

Theorem 2.5 (van der Waerden) *Given a function $f: \mathbb{N} \rightarrow [N]$, there exists $1 \leq i \leq N$ such that*

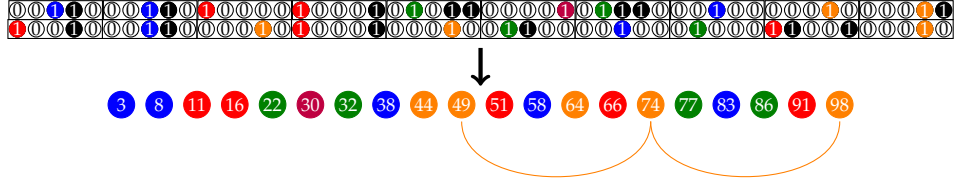
$$\forall k \in \mathbb{N}, \exists r, \ell \in \mathbb{N}: f(r) = f(r + \ell) = \dots = f(r + k\ell) = i.$$

Such a function f is often referred to as a *colouring*, with the analogy that each integer in the image represents a colour. In this context, van der Waerden's theorem is often phrased as "for a colouring of $\mathbb{N}$, one can find arbitrarily long monochromatic arithmetic progressions". Note also that it does not make a difference if we order the quantifiers as $\exists i: \forall k \in \mathbb{N}$ and $\forall k \in \mathbb{N}: \exists i$ - in other words, the properties of (i) having arbitrarily long monochromatic A.P.s and (ii) having some given colour with arbitrarily long monochromatic A.P.s are equivalent.

With Theorem 2.5, the proof of Lemma 2.4 is quite straightforward:

Proof. (Lemma 2.4) By Definition 2.3, a set S is syndetic if and only if there exists some N such that any group of N consecutive integers contains an element of S . Consider now a syndetic set S and a valid choice of N for that set.

Let $B_i = \{(i-1)N + 1, (i-1)N + 2, \dots, iN\}$. We know by syndeticity that $B_i \cap S \neq \emptyset, \forall i \in \mathbb{N}$. For every i let $S(i)$ be an element of S in B_i (if there are multiple such elements, take any one of them) and consider the function f defined by $f(i) = S(i) \bmod N$. This function defines a colouring and in particular by Theorem 2.5 there is some $j \in [N]$ such that there are arbitrarily long arithmetic progressions mapped to j . Consider such an A.P. $r, r + \ell, \dots, r + k\ell$. Then it follows by the definition of f that $\{ir + j, i(r + \ell) + j, \dots, i(r + k\ell) + j\} \subset S$. But this is an A.P. of the same length and in particular we can find arithmetic progressions of arbitrary length in S . $\square$



Note that many results like Lemma 2.4 admit both "finitistic" and "infinitistic" versions. For example, consider the following.

Lemma 2.6 *The following assertions are equivalent.*

- (i) *Syndetic sets contain arbitrarily long arithmetic progressions.*
- (ii) *For any $N, k \in \mathbb{N}$, there is some threshold $M = M(N, k) \in \mathbb{N}$ such that any syndetic set S where the difference between consecutive elements in S is bounded by N , has the property that $S \cap [M]$ contains an arithmetic progression of length k .*

Proof. (ii) $\implies$ (i): This is trivial.

(i) $\implies$ (ii): We will show the contrapositive. Assume there are $N, k \in \mathbb{N}$ for which no such M exists. It follows that for all $M \in \mathbb{N}$, there are syndetic sets with gap size N that contain no arithmetic progressions when taking their intersection with $[M]$. Consider now syndetic sets with this property for every $M = 1, 2, \dots$, and call them $T_1, T_2, \dots$ respectively, and for $i \in \mathbb{N}$ let $S_i = T_i \cap [i]$. Let $\mathcal{S} = \{S_1, S_2, \dots\}$. Let $B_i = \{(i-1)N + 1, (i-1)N + 2, \dots, iN\}$.

We know by syndeticity that $B_i \cap S_j \neq \emptyset, \forall i \in \mathbb{N}, j \in \mathbb{N} > iN$. Consider first the intersections $B_1 \cap S_{N+1}, B_1 \cap S_{N+2}, \dots$ - there must be a number d in $B_1 = \{1, 2, \dots, N\}$ which appears infinitely many times in these intersections. Pick this number to be the first element of a new set S , and let

$$S_1 = \{S_i \in \mathcal{S} : d \in S_i\}$$

and note that $\mathcal{S}_1$ nonempty contains infinitely many sets. Considering B_2 now, we again consider some integer that appears infinitely many times in $\mathcal{S}_1$, add it to S , and define $\mathcal{S}_2$ analogously to before. Note that at every step, we have that

- $\mathcal{S}_i$ is infinite.
- S is a subset of every element of $\mathcal{S}_i$.

Continuing this way, the set S we construct does not contain any arithmetic progressions of length k , for if it did, then it would suffice to consider the largest element N of this progression; at the moment in our process that N was added to S , S was a subset of some $S_M, M > N$ which does not have arithmetic progressions of length k , contradiction. Furthermore, S is syndetic - it is easy to see the difference of consecutive elements is at most $2N$. $\square$

When discussing a family of sets, one often considers the sets intersecting with the elements of the family. Let $\mathcal{P}(\mathbb{N})$ denote the set of all subsets of $\mathbb{N}$.

Definition 2.7 (Dual family of a family of sets) For a family $\mathcal{F} \subseteq \mathcal{P}(\mathbb{N})$, the dual family $\mathcal{F}^*$ is defined as follows:

$$\mathcal{F}^* := \{B \in \mathcal{P}(\mathbb{N}) : \forall A \in \mathcal{F}, A \cap B \neq \emptyset\}.$$

In other words, the dual family consists of all sets intersecting with every single element of $\mathcal{F}$. In the case of syndetic sets, the dual family has a very particular form.

Definition 2.8 (Thick set) A set $A \subseteq \mathbb{N}$ is thick if it contains arbitrarily long sequences of consecutive integers.

It is easy to show that the family of thick sets is the dual family of the family of syndetic sets and vice versa (note that in general it is not necessarily true that the dual family of the dual family should be the original family). The sets obtained by taking the intersection of a syndetic and thick set have a name of their own.

Definition 2.9 (Piecewise syndetic set) A set A is piecewise syndetic if there exists a syndetic set S and a thick set T such that $A = S \cap T$.

Lemma 2.10 Piecewise syndetic sets contain arbitrarily long arithmetic progressions.

This is a consequence of (ii) in Lemma 2.6 and is left as an exercise for the reader.

Finally, we come to notion of largeness for families of subsets of $\mathbb{N}$.

Definition 2.11 (Upward closure) A family of sets $\mathcal{F} \subseteq \mathcal{P}(\mathbb{N})$ is upwards closed if for any set $F \in \mathcal{F}$, we have that if $F \subset F' \implies F' \in \mathcal{F}$.

It is easy to see that the families of syndetic sets, thick sets and piecewise syndetic sets are all upwards closed.

Definition 2.12 (Partition regularity) *A family of sets $\mathcal{F} \subseteq \mathcal{P}(\mathbb{N})$ is partition regular if for any set $F \in \mathcal{F}$, we have that if*

$$F = F_1 \cup F_2 \cup \cdots \cup F_N$$

is a disjoint union (i.e., for $1 \leq i < j \leq N$, $F_i \cap F_j = \emptyset$) then there exists $n \in [N]$ such that $F_n \in \mathcal{F}$.

Partition regularity is often called the *Ramsey property*.

Thick sets are clearly not partition regular; splitting a thick set into its even and odd elements yields two sets, neither of which is thick. The same is true for syndetic sets: it suffices to assign the first element to one set, the next two to the second, the next three to the first set, and so forth; this will yield two sets with unbounded gaps.

Theorem 2.13 *The family of piecewise syndetic sets is partition regular.*

To prove this, we will prove a more general result on the intersection of a family and its dual family.

Lemma 2.14 *Given an upwards closed family of sets $\mathcal{F}$ and its dual $\mathcal{F}^*$, the family $\mathcal{G}$ defined as*

$$\mathcal{G} := \{F \cap F' : F \in \mathcal{F}, F' \in \mathcal{F}^*\}$$

is partition regular.

Proof. Note that it is clearly enough to prove partition regularity for splitting into two sets, as by induction it will hold for any finite number of sets. Consider families $\mathcal{F}, \mathcal{F}^*, \mathcal{G}$ as defined in Lemma 2.14, and let $G = F \cap F'$ be an element of $\mathcal{G}$. Note that $\mathcal{F}^*$ will also trivially be upwards closed. In particular, this implies that $\mathcal{F}^{**} = \mathcal{F}$; this is left to the reader to verify (the proof hinges on the fact that in the general case $\mathcal{F}^{**} \subset \mathcal{F}$ and any "missing" sets appear via upwards closure.)

We split $G = G_1 \cup G_2$ into disjoint sets and take $H := G_1 \cup (F \setminus G)$. In particular, $B = H \cup F'$, so if $H \in \mathcal{F}$ it follows that $G_1 \in \mathcal{G}$ as desired. If this is not the case, then the complementary set $H^c \in \mathcal{F}^*$ (if H is not in $\mathcal{F}$ then some subset of H^c is in $\mathcal{F}^*$ - recall that $\mathcal{F}^{**} = \mathcal{F}$. Then upwards closure implies $H^c \in \mathcal{F}^*$). However, we now have $G_2 = F \cap H^c$ and we again conclude. $\square$

Lemma 2.14 implies Theorem 2.13 via the fact that thick sets are the dual of syndetic sets. Theorem 2.13 along with Lemma 2.10 implies Theorem 2.5. This is of course not a proof of van der Waerden's theorem, as we assumed Theorem 2.5 held in order to prove Lemma 2.4, but it is actually possible to prove Lemma 2.4 without first assuming that Theorem 2.5 is true (although we will not show any such alternate proof).

2.2 Measure-preserving systems and recurrence

We begin this subsection with two basic definitions.

Definition 2.15 (Measure-preserving system) A measure-preserving system $(X, \mathcal{B}, \mu, T)$ is a probability space $(X, \mathcal{B}, \mu)$ equipped with a measure-preserving transformation $T: \mathcal{B} \rightarrow \mathcal{B}$, meaning $\forall A \in \mathcal{B}, \mu(T^{-1}(A)) = \mu(A)$.

Definition 2.16 (Ergodic measure-preserving system) A measure-preserving system $(X, \mathcal{B}, \mu, T)$ is called ergodic if it has the following property: For any set $A \in \mathcal{B}$ satisfying

$$\mu(A \Delta T^{-1}A) = 0$$

we have $\mu(A) \in \{0, 1\}$.

We restate the key results from the introduction here. The first result is due to Poincaré[1].

Theorem 2.17 (Poincaré recurrence theorem) Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. For any $A \in \mathcal{B}$ with $\mu(A) > 0$, we have infinitely many $n \in \mathbb{N}$ such that $\mu(A \cap T^{-n}A) > 0$.

Proof. We will prove a stronger result; namely that for all $n \in \mathbb{N}$, there exists a increasing sequence of positive integers $(j_i)_{i=1}^n \in \mathbb{N}$ such that

$$\mu(A \cap T^{-j_1}A \cap \dots \cap T^{-j_n}A) > 0.$$

Let $N_1 := \lfloor \frac{1}{\mu(A)} \rfloor$; it follows that the $N_1 + 1$ sets $A, T^{-1}A, \dots, T^{-N_1}A$ cannot all have pairwise intersections with measure zero, as their total measure is greater than 1. So there exists $0 \leq i < j \leq N_1 + 1$ such that $\mu(T^{-i}A \cap T^{-j}A) > 0$. Then, since

$$\mu(A \cap T^{-(j-i)}A) = \mu(T^{-i}(A \cap T^{-(j-i)}A)) = \mu(T^{-i}A \cap T^{-j}A) > 0 \quad (2.1)$$

it follows that we have $k_1 := j - i \in \mathbb{N}$ such that $A \cap T^{-k_1}A$ has positive measure. Now, by letting $N_2 := \lfloor \frac{1}{\mu(A \cap T^{-k_1}A)} \rfloor$ and repeating the previous argument with the $N_2 + 1$ sets $A \cap T^{-k_1}A, T^{-1}A \cap T^{-k_1-1}A, \dots, T^{-N_2}A \cap T^{-N_2-k_1}A$, we can find k_2 such that

$$\mu((A \cap T^{-k_1}A) \cap (T^{-k_2}A \cap T^{-k_2-k_1}A)) > 0,$$

giving us at least two elements for our sequence (k_1 and $k_2 + k_1$ are distinct). Repeating in this manner, we can show the desired result. $\square$

There is another formulation of the recurrence theorem sometimes used. We state it here; we give a proof from Frantzikinakis and McCutcheon's survey on recurrence[13].

Theorem 2.18 (Poincaré recurrence theorem, II) *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. For any $A \in \mathcal{B}$ with $\mu(A) > 0$, the set of points $x \in A$ such that there exists $N \in \mathbb{N}$ with $T^{-n}(x) \notin A$ for all $n > N$ has measure 0.*

Proof. Let B be the set of all $x \in A$ such that $T^{-n}x \notin A$ for all $n \in \mathbb{N}$. This set is measurable as it is the union of all the $T^{-n}A$ subtracted from A . This implies in turn that $B, T^{-1}B, T^{-2}B, \dots$ are all disjoint as if there exist $i < j \in \mathbb{N}$ such that $T^{-i}B \cap T^{-j}B \neq \emptyset$, it follows that $B \cap T^{-(j-i)}B \neq \emptyset$, contradicting the definition of B . However, since we now have

$$1 \geq \mu \left(\bigcup_{i=0}^{\infty} T^{-i}B \right) = \sum_{i=0}^{\infty} \mu(T^{-i}B)$$

but all the terms in the RHS have the same measure, we must have $\mu(B) = 0$. So for almost every x , one has $T^{-n}(x) \in A$ for some $n \in \mathbb{N}$. We can repeat this argument, replacing T with T^{N+1} , to conclude that one has $T^{-n(N+1)}(x) \in A$ for some $n \in \mathbb{N}$, and $n(N+1) > N$. $\square$

Finally, we state Furstenberg's correspondence principle (see Bergelson's papers[14, 15, 16]).

Theorem 2.19 (Furstenberg's correspondence principle) *Let $E \subseteq \mathbb{N}$. There exists a measure preserving system $(X, \mathcal{B}, \mu, T)$ and a set $S \in \mathcal{B}$ with $\mu(S) = \bar{d}(E)$ such that for any $k \in \mathbb{N}$ and $n_1, \dots, n_k \in \mathbb{N}$, we have*

$$\mu(A \cap T^{-n_1}A \cap \dots \cap T^{-n_k}A) \leq \bar{d}(E \cap (E - n_1) \cap \dots \cap (E - n_k)).$$

We will not prove Furstenberg's correspondence principle. A proof can be found as Theorem 4.17 in Bergelson's 1996 paper[15], which gives a general form of Furstenberg's correspondence principle for amenable groups.

2.3 Harmonic analysis

We now state a few important results from Fourier analysis.

Definition 2.20 (Fourier coefficients) *Consider a periodic function $f \in L^1(\mathbb{R})$, with period 1. We use the well-known shorthand*

$$e(x) := e^{2i\pi x}$$

We then define, for $m \in \mathbb{Z}$:

$$\hat{f}(m) = \int_{\mathbb{T}} f(x)e(-mx)dx. \quad (2.2)$$

We then have

$$f(x) = \sum_{m=-\infty}^{\infty} \hat{f}(m)e(mx).$$

The elements of the sequence $(\hat{f}(m))_{m=-\infty}^{\infty}$ are called the Fourier coefficients of f .

Definition 2.21 (Fourier coefficients of a measure on the torus) Consider a Borel measure μ on the torus $\mathbb{T}$. We then define, for $m \in \mathbb{Z}$:

$$\widehat{\mu}(m) = \int_{\mathbb{T}} e(-mx) d\mu(x).$$

The elements of the sequence $(\widehat{\mu}(m))_{m=-\infty}^{\infty}$ are called the Fourier coefficients of μ .

These definitions are analogous in that a Borel measure on the torus can be extended to a periodic, L^1 function on $\mathbb{R}$.

It is standard to call functions $f(x)$ expressed in the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^M a_n \cos(2\pi nx) + \sum_{n=1}^M b_n \sin(2\pi nx).$$

(where $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are complex-valued sequences) *trigonometric polynomials*. All trigonometric polynomials are of course periodic with period 1. Furthermore, if we apply Euler's formula ($e(x) = \cos(2\pi x) + i \sin(2\pi x)$), then defining

$$c_m = \begin{cases} a_0, & m = 0 \\ \frac{1}{2}(a_m - ib_m), & m > 0 \\ \overline{c_{-m}}, & m < 0. \end{cases}$$

we can express the trigonometric polynomials in the form

$$f(x) = \sum_{m=-M}^M c_m e(mx). \quad (2.3)$$

When we will henceforth refer to trigonometric polynomials, we write them in the second form. Note as mentioned previously that $c_m = \overline{c_{-m}}$ for the coefficients when written in the exponential form. The set of c_m not equal to 0 for positive m is called the *spectrum* of the trigonometric polynomial. Of course, $(c_m)_{m=-M}^M$ is equal to the sequence of Fourier coefficients $(\widehat{f}(m))_{m=-M}^M$ of the trigonometric polynomial.

If we have a function $f(x)$ written in the form in (2.3) but we do not necessarily have $c_m = \overline{c_{-m}}$, we call it an *exponential polynomial* instead. We can define the Fourier coefficients of an exponential polynomial similarly to (2.2), just now taking

$$\widehat{f}(m) = \int_{-\infty}^{\infty} f(x) e(-mx) dx.$$

and we will get $\widehat{f}(m) = c_m$.

2.4 Equidistribution and the van der Corput theorem

Definition 2.22 (Equidistribution) Let $(x_n)_{n=1}^{\infty}$ be a sequence taking values in $[0, 1]$. We say it is equidistributed on the interval $[0, 1]$ if for every subinterval $[a, b] \subset [0, 1]$ we have

$$\lim_{n \rightarrow \infty} \frac{|\{x_1, \dots, x_n\} \cap [a, b]|}{n} = b - a.$$

There is of course an analogous definition for being equidistributed on other intervals, but we will only be interested in $[0, 1]$. Consequently, when we say from now on that a sequence is equidistributed, it is implied that we are talking about $[0, 1]$ and that the sequence takes values on this interval.

The following lemma provides a useful equivalent form of the notion of equidistribution.

Lemma 2.23 (Riemann integral criterion for equidistribution) Let $(x_n)_{n=1}^{\infty}$ be a sequence of real numbers. Then the following two conditions are equivalent:

- (i) $(x_n)_{n=1}^{\infty}$ is equidistributed.
- (ii) For every Riemann-integrable function $f: [0, 1] \rightarrow \mathbb{C}$ we have that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(x_n) = \int_0^1 f(x) dx \quad (2.4)$$

We will not provide the proof in full, but sketch the main ideas. A key observation is that if we change (ii) from all Riemann-integrable functions to just indicator functions of intervals, the equivalence is trivial, as it corresponds exactly to the definition of equidistribution. Therefore it remains to show (ii) for interval indicator functions implies (ii) for an arbitrary complex-valued function.

Since (2.4) is preserved by the addition of functions, we can go from indicator functions to step functions. Going from step functions to arbitrary real-valued Riemann functions is just a standard epsilon-delta argument, using Darboux's result that if the liminf of integrals of step functions above the function and the limsup of integrals of step functions below the function are equal, these two values are equal to the Riemann integral. To go from real-valued to complex-valued integrable functions is a consequence of the fact that we can write any complex integrable function as the sum of a purely integrable real and purely imaginary integrable function, and dividing the latter case by the imaginary constant i gives the real case again.

Of course, many sequences are not equidistributed in the interval $[0, 1]$ but may be when you look at them modulo 1 - for instance, the sequence $\pi, 2\pi, 3\pi, \dots$ - which leads to a natural adjustment of the definition.

Definition 2.24 (Equidistribution modulo 1) A sequence $(x_n)_{n=1}^{\infty}$ of reals is said to be equidistributed modulo 1 if the sequence $(x_i - \lfloor x_i \rfloor)_{n=1}^{\infty}$ is equidistributed.

Going forward, when we say a sequence is equidistributed, it will be implicit that we are considering it modulo 1 (on top of the fact that it is already implicit that the interval on which it is equidistributed is $[0, 1]$).

Lemma 2.25 For any $x \in \mathbb{R}$, $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $nx \bmod 1 < \varepsilon$.

Proof. If x is rational then then writing $x = \frac{a}{b}$ with $a \in \mathbb{Z}, b \in \mathbb{N}$ one gets that $bx \in \mathbb{N} = 0 \bmod 1$ and we are done.

Suppose now that x is irrational. Notice that the sequence cannot ever contain the same term twice (if $\exists a, b \in \mathbb{N}$ such that $ax = bx \bmod 1$ then $(a - b)x \in \mathbb{N}$, contradiction) and consider $\varepsilon > 0$. By the pigeonhole principle, we must have two terms ix, jx with $i < j$ such that $|ix \bmod 1 - jx \bmod 1| < \varepsilon$ (simply partition $[0, 1]$ into subintervals of size at most ε ; there must be two elements of $x, 2x, \dots$ in the same subinterval). But then $|i - j|x \bmod 1 < \varepsilon$ and it suffices to take $n = j - i$ to either have that $nx \bmod 1 < \varepsilon$ (and we are done) or $nx \bmod 1 > 1 - \varepsilon$. In the latter case, if we consider now $nx, 2nx, \dots \bmod 1$, it is a $(1 - nx \bmod 1)$ -dense subset of $[0, 1]$ and contains a term in the interval $[0, \varepsilon]$. $\square$

In particular, the proof of Lemma 2.25 also shows that if x is irrational, for any interval $I \subseteq [0, 1]$, there are infinitely many $n \in \mathbb{N}$ such that $nx \bmod 1 \in I$. Indeed, since for any ε there exists m such that $0 < mx \bmod 1 < \varepsilon$, then the multiples $mx, 2mx, \dots \bmod 1$ form an ε -dense subset of $[0, 1]$, and we can find arbitrarily many terms in any given interval by taking ε sufficiently small.

Now we note a key result in equidistribution proved by Weyl[17] in 1916.

Lemma 2.26 (Weyl's criterion) Suppose $(x_n)_{n=1}^{\infty}$ is a sequence taking values in $[0, 1]$. Then the following two conditions are equivalent:

- (i) $(x_n)_{n=1}^{\infty}$ is equidistributed.
- (ii) For all nonzero integers ℓ we have that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e(\ell x_n) = 0.$$

Recall that the notation $e(t)$ is defined in Definition 2.20. We do not prove this result, but the fact that (i) implies (ii) is an application of Lemma 2.23, and the other direction of the proof uses Stone-Weierstrass to approximate continuous functions with trigonometric polynomials and then finish similarly to the proof sketch of 2.23.

2.4. Equidistribution and the van der Corput theorem

One of the most famous results on equidistributed sequences is the following theorem of van der Corput from 1931[7].

Theorem 2.27 (van der Corput's difference theorem) *Let $(x_n)_{n=1}^{\infty}$ be a sequence such that for all $h \in \mathbb{N}$ the sequence $(x_{n+h} - x_n)_{n=1}^{\infty}$ is equidistributed. Then the sequence (x_n) is itself equidistributed.*

A proof of Theorem 2.27 can be found, for example, in a paper of Bergelson[15] that we show in Section 3.2.

Chapter 3

Sets of recurrence and van der Corput sets

In this chapter, we turn our attention to some standard families of sets that are important in ergodic Ramsey theory. Our aim is to understand the relationships between these families and survey some notable results.

In particular, we will discuss both van der Corput sets and sets of recurrence, with the main goal of providing a thorough exposition of Mountakis'[10] proof of the existence of a set of strong recurrence that is not van der Corput.

3.1 Sets of recurrence

We begin by providing a definition sets of recurrence, a concept which was introduced by Furstenberg (see [2, 18]).

Definition 3.1 (Set of recurrence) *A set $R \subseteq \mathbb{N}$ is a set of recurrence if for any measure-preserving systems $(X, \mathcal{B}, \mu, T)$ and all $A \in \mathcal{B}$ with $\mu(A) > 0$, there exists $n \in R$ such that $\mu(A \cap T^{-n}A) > 0$.*

By Theorem 2.17, we already know $\mathbb{N}$ is a set of recurrence. To prove Theorem 2.17, we used the pigeonhole principle; a similar argument can also be used to find other examples of a set of recurrence. Given a measure-preserving system $(X, \mathcal{B}, \mu, T)$, we already know for any set $A \in \mathcal{B}$ with $\mu(A) > 0$, we cannot have an infinite increasing sequence of positive integers $n_1, n_2, \dots$ such that all the sets of the form $T^{-n_i}A$ all have pairwise intersection with measure 0, as the total measure of X is finite. This immediately implies that for any infinite set $S \subseteq \mathbb{N}$, the set $S - S$ is a set of recurrence, as if $T^{-n_j}A$ and $T^{-n_i}A$ do not have intersection with measure 0 (WLOG $j > i$), then A and $T^{-(n_j - n_i)}A$ do not have intersection with measure 0, analogously to (2.1).

3.1.1 Intersectivity

Now we arrive at our main result about sets of recurrence. It turns out that Definition 3.1 is equivalent to each of the two definitions below.

Definition 3.2 (Set of combinatorial recurrence) A set $S \subseteq \mathbb{N}$ is an set of combinatorial recurrence if for any set $E \subseteq \mathbb{N}$ such that $\bar{d}(E) > 0$, there exists $n \in S$ such that $\bar{d}(E \cap (E - n)) > 0$.

Definition 3.3 (Intersective set) A set $S \subseteq \mathbb{N}$ is an intersective set if for any set $E \subseteq \mathbb{N}$ such that $\bar{d}(E) > 0$, there exist $n \in S$ such that $E \cap (E - n) \neq \emptyset$.

Note that a set $S \subseteq \mathbb{N}$ is intersective if and only if for any E with $\bar{d}(E) > 0$, $(E - E) \cap S \neq \emptyset$; we leave this fact as an exercise to the reader.

It seems natural that Definition 3.2 and Definition 3.3 should be closely linked; indeed, the fact that combinatorial recurrence implies intersectivity is immediate. That it should be an equivalence is altogether less obvious, and the fact that "standard" recurrence coincides with both will be our next result.

Theorem 3.4 The following are equivalent, for a set $S \subseteq \mathbb{N}$.

- (i) S is a set of recurrence.
- (ii) S is a set of combinatorial recurrence.
- (iii) S is an intersective set.

Proof. (i) $\implies$ (ii): This is just a direct application of Furstenberg's correspondence principle (Theorem 2.19.)

(ii) $\implies$ (iii): This is trivial - $\bar{d}(E \cap (E - n)) > 0$ clearly implies that $E \cap (E - n)$ is nonempty.

(iii) $\implies$ (i): We will use a result from a paper of Bergelson[6]. Namely, we have the following as Theorem 1.2 from that paper.

Theorem 3.5 Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and let $A \in \mathcal{B}$ such that $\mu(A) = a > 0$. Then there exists a increasing sequence $(x_n)_{n=1}^\infty$ such that if $E = \{x_1 < x_2 < x_3 \dots\}$ then $d(E) > a$ and $\forall m \in \mathbb{N}$,

$$\mu(A \cap T^{-x_1} A \cap \dots \cap T^{-x_m} A) > 0. \quad (3.1)$$

Proof. (Theorem 3.5) We will prove a slightly more general result. Rather than showing that one can find a sequence $(x_n)_{n=1}^\infty$ with the property in (3.1), we will replace the sets $A, T^{-1}A, \dots$ with arbitrary sets $A_1, A_2, \dots$ with measure a . In other words, for any sequence of sets $(A_i)_{i=1}^\infty$ in $\mathcal{B}$, $\mu(A_i) = a$, we have an increasing sequence $E \subseteq \mathbb{N}$ such that if $E = \{x_1 < x_2 < x_3 < \dots\}$, $d(E) > a$ and for all $m \in \mathbb{N}$,

$$\mu(A_{x_1} \cap A_{x_2} \cap \dots \cap A_{x_m}) > 0. \quad (3.2)$$

Notice that if (3.2), then we have that for any finite subset $F \subseteq E$,

$$\mu(\cap_{i \in F} A_i) > 0 \quad (3.3)$$

and of course (3.3) also implies (3.2). So we proceed to a proof showing (3.3) is possible. For any nonempty finite set $F \subseteq \mathbb{N}$, let $A_F := \mu(\cap_{i \in F} A_i)$. We begin by showing there exists a set $B \in \mathcal{B}$ of measure 0 such that $(X \setminus B) \cap A_F \neq \emptyset$, implies $\mu(A_F) > 0$. Note that $\mathbb{1}_{A_F} := \prod_{i \in F} \mathbb{1}_{A_i}$. Then, we define

$$\begin{aligned} B_F &:= \{t \in X : \mathbb{1}_{A_F}(t) > \sup \text{ess}|f|\} \\ B &:= \bigcup_{F \in \mathbb{N}, |F| < \infty} B_F. \end{aligned}$$

For B_F to be nonempty, since $\mathbb{1}_{A_F}(t)$ is either equal to 0 or to 1, it follows that $\mu(A_F) = 0$ but $A_F \neq \emptyset$. B clearly has measure 0 as the union of countably many sets of measure 0. Now, take any A_F where $(X \setminus B) \cap A_F \neq \emptyset$. By the previous remark, $\mu(A_F)$ cannot be 0 as A_F is nonempty. So we have proven the set B has the desired properties.

Hereafter, we consider only sets A_F in $(X \setminus B)$. We now need to show that there exists an infinite set $E \subseteq \mathbb{N}$ such that for any finite subset $F \subseteq E$,

$$(A_F \setminus B) \neq \emptyset. \quad (3.4)$$

Define $f_N(t) := \frac{1}{N} \sum_{n=1}^N \mathbb{1}_{A_n}(t)$; which is a function taking values in $[0, 1]$ and whose integral $\int_X f_N d\mu = a > 0$. Define as well

$$f(t) = \limsup_{N \rightarrow \infty} f_N(t).$$

Fatou's lemma tells us that the integral $\int_X f d\mu = \int_{X \setminus B} f d\mu$ is at least a . Consequently, there is some $u \in X \setminus B$ where $f(u) \geq a$. If we now take $E := \{n \in \mathbb{N} : u \in A_n\}$ then $\bar{d}(E) = \limsup f(u) \geq a$. In particular, as $u \in X \setminus B$, E satisfies (3.4) and hence (3.3). $\square$

With this result, the implication becomes straightforward to prove. Let S be an interseptive set and $(X, \mathcal{B}, \mu, T)$ be any measure-preserving system. Let $(x_n)_{n=1}^\infty$ and E be as described in the theorem. As $\bar{d}(E) > 0$, we know that $\exists i < j \in \mathbb{N}$ such that $x_j - x_i \in S$. But then it follows from (4.1) that

$$\mu(T^{-x_i} A \cap T^{-x_j} A) > 0$$

implying that, as T is measure-preserving,

$$\mu(A \cap T^{-(x_j - x_i)} A) > 0$$

and so as $x_j - x_i \in S$, S is a set of recurrence. This completes the proof that (iii) $\implies$ (i).

Bertrand-Mathis[19] provided another proof, assuming T is invertible, which we show here as this proof has not been previously translated to English. Readers who are satisfied with the proof already presented may skip ahead to 3.1.2.

We need two lemmas. The first is easy to prove:

Lemma 3.6 *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and consider any $A \in \mathcal{B}$ such that $\mu(A) > 0$. Let $W_A \subseteq \mathbb{Z}^2$ be defined as*

$$W_A := \{n, m \in \mathbb{Z} : \mu(T^{-n}A \cap T^{-m}A) = 0\}.$$

Then the set

$$V_A := \bigcup_{(n,m) \in W_A} (T^{-n}A \cap T^{-m}A)$$

is T -invariant and has measure 0.

Proof. (Lemma 3.6) As the countable union of sets of measure 0, V_A is clearly of measure 0. Being T -invariant is a natural consequence of the fact that $\mu(T^{-n}A \cap T^{-m}A) = \mu(T^{-i}(T^{-n}A \cap T^{-m}A)) = \mu(T^{-n-i}A \cap T^{-m-i}A)$, since T is measure-preserving. $\square$

Before proving the second lemma, we need the following auxiliary result by Khintchine.

Theorem 3.7 (Khintchine Recurrence Theorem) *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and consider any $A \in \mathcal{B}$ with $\mu(A) > 0$. For every $\varepsilon > 0$ the set*

$$\{n \in \mathbb{N} : \mu(A \cap T^n A) > \mu(A)^2 - \varepsilon\}$$

is syndetic.

We will not prove the theorem here; a proof can be found in a survey of Bergelson[20].

Lemma 3.8 *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and consider any $A \in \mathcal{B}$ such that $\mu(A) > 0$. Call an element $x \in A$ focused if we can find two sequences, $(x_i)_{i=1}^\infty$ of elements in A and $(n_i)_{i=1}^\infty$ of positive integers, such that the union of the n_i is a set with strictly positive upper density, and that for all $i \in \mathbb{N}$*

$$x = T^{-n_i}(x_i).$$

The set F_A of all focused elements has nonzero measure.

Proof. (Lemma 3.8) Invoking Khintchine, we know that there is a sequence of integers m_i whose union is syndetic (and therefore has positive density d) such that

$$\mu(T^{-m_i}A \cap A) \geq \frac{\mu(A)^2}{2}$$

simply by taking $\varepsilon = \frac{\mu(A)^2}{2}$ in the statement of Theorem 3.7. Let f_n be the indicator function of the set $T^n A \cap A$; in particular, the measure of $T^n A \cap A$ is exactly the integral of this function w.r.t $d\mu$. It follows that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum \int_X f_n d\mu \geq \frac{d}{2} (\mu(A))^2$$

as the proportion of terms contributing a measure of at least $\frac{\mu(A)^2}{2}$ is d , and we get the desired inequality by taking limits.

Now, note that $x \in F_A$ is equivalent to

$$\limsup_{N \rightarrow \infty} \frac{1}{N} |\{n \in [N] : x \in T^{-n} A \cap A\}| > 0$$

as a focused element x lies in A and also in $T^{-n_i} A$ for a set of n_i having upper positive density. In particular

$$\limsup_{N \rightarrow \infty} \frac{1}{N} |\{n \in [N] : x \in T^{-n} A \cap A\}| = \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{1 \leq n \leq N} f_n(x)$$

and so we just need to prove the set D_A , which we define as the set of x where the latter expression is greater than 0, has positive measure. Let us instead focus on the set of x *not* satisfying the latter condition - namely, $x \in X \setminus D_A$ - and prove that $\mu(X \setminus D_A) < 1$.

For the sake of contradiction, we therefore assume the set of $x \in X \setminus D_A$; i.e., the x satisfying $\limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{1 \leq n \leq N} f_n(x) = 0$, has measure 1.

It follows now that since

$$\frac{1}{N} \sum_{1 \leq n \leq N} \int_X f_n d\mu = \frac{1}{N} \sum_{1 \leq n \leq N} \int_{X \setminus D_A} f_n d\mu \quad (3.5)$$

we can apply a limit to the RHS expression in (3.5) and use Lebesgue's dominated convergence theorem to swap the limit and the integral as follows:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{1 \leq n \leq N} \int_{X \setminus D_A} f_n d\mu = \int_{X \setminus D_A} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{1 \leq n \leq N} f_n d\mu$$

which is 0 by the definition of $X \setminus D_A$. In particular, this contradicts the inequality we derived due to Khintchine. $\square$

We can finally now proceed to the proof that intersectivity implies recurrence.

Proof. (3. $\implies$ 1.) Let S be an intersective set, $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and consider any $A \in \mathcal{B}$ such that $\mu(A) > 0$. We will show the implication for the case where T is invertible. We wish to show

that there exists $n \in S$ such that $\mu(T^{-n}A \cap A) > 0$. Take V_A as defined in Lemma 3.6 and F_A as defined in Lemma 3.8, and take any $x \in F_A \setminus V_A$, which is nonempty as it has positive measure. Then take two sequences, $(x_i)_{i=1}^\infty$ of elements in A and $(n_i)_{i=1}^\infty$ of positive integers as defined in Lemma 3.8. Consequently, as the union of all the n_i has positive upper density, and S is presumed to be intersective, we know that there exist n_j, n_k such that

$$n_j - n_k \in S, x = T^{-n_j}(x_j) = T^{-n_k}(x_k)$$

and with the invertibility of T , one has $x_j = T^{-(n_j - n_k)}(x_k)$. In particular, $T^{-(n_j - n_k)}A \cap A$ contains x_j , and $x_j \notin D_A$ as D_A is T -invariant and does not contain x either. By definition of D_A , it follows that $\mu(T^{-(n_j - n_k)}A \cap A)$ cannot be 0 and S is a set of recurrence as desired. $\square$

This concludes the proof of Theorem 3.4. $\square$

3.1.2 Partition regularity

Finally, we can show that sets of recurrence are partition regular. We present here a proof from another survey by Bergelson[15].

Theorem 3.9 *The family of sets of recurrence is partition regular. In other words, given any set of recurrence $R \subseteq \mathbb{N}$, if we write R as the disjoint union of finitely many sets $R = R_1 \cup R_2 \cup \dots \cup R_k$, then at least one of the sets R_i is a set of recurrence.*

Proof. Assume otherwise, meaning that there exists some set of recurrence $R = R_1 \cup R_2 \cup \dots \cup R_k$ where each R_i for $1 \leq i \leq k$ is not a set of recurrence. Then, there exist measure-preserving systems $(X_i, \mathcal{B}_i, \mu_i, T_i)$ corresponding to each R_i respectively where we can find $A_i \in \mathcal{B}_i, \mu(A_i) > 0$ such that $\mu_i(A_i \cap T_i^{-n}A_i) = 0$ for any $n \in R_i$. We will utilise the product measure-preserving system $(X, \mathcal{B}, \mu, T)$, where

$$X = X_1 \times \dots \times X_k;$$

$$\mathcal{B} \text{ is the } \sigma\text{-algebra generated by } \mathcal{B}_1 \times \dots \times \mathcal{B}_k;$$

$$\mu \text{ is the product measure;}$$

$$T \text{ is the product transformation.}$$

In particular, as R is a set of recurrence, we know R must intersect with the set of returns for any element of $\mathcal{B}$, notably $A = A_1 \times \dots \times A_k \in \mathcal{B}$. This implies that for some $n \in R$,

$$\mu(A \cap T^{-n}A) = \mu_1(A_1 \cap T_1^{-n}A_1) \cdots \mu_k(A_k \cap T_k^{-n}A_k) > 0$$

implying $\mu_i(A_i \cap T_i^{-n}A_i) > 0$ for all $1 \leq i \leq k$, which contradicts the assumptions made about the sets R_j . $\square$

3.2 Van der Corput sets

Theorem 2.27 relates the equidistribution of the sequences, $(x_{n+h} - x_n)_{n=1}^{\infty}$, $h \in N$ to the equidistribution of $(x_n)_{n=1}^{\infty}$. A natural question to enquire is whether we actually need all $h \in \mathbb{N}$, or whether there exists $H \subseteq \mathbb{N}$ such that checking only for $h \in H$ suffices.

Definition 3.10 (van der Corput set) *Let H be a set with the following property: for any sequence $(x_n)_{n=1}^{\infty}$ such that for all $h \in H$, the sequence $(x_{n+h} - x_n)_{n=1}^{\infty}$ is equidistributed, the sequence $(x_n)_{n=1}^{\infty}$ is itself equidistributed. We call H a van der Corput set.*

It will be of interest in this section to identify examples of van der Corput sets, as well as necessary and sufficient properties of a set for it to be van der Corput.

3.2.1 Van der Corput's difference theorem

Let us start this section with a proof of a more general variant of Theorem 2.27 for Hilbert spaces, as given in a survey of Bergelson[15]. For more details, one can also consult of the proof in a survey of Bergelson and Moreira[21].

Theorem 3.11 *If $(u_i)_{i=1}^{\infty}$ is a bounded sequence in a Hilbert space $\mathcal{H}$ and if for any $h \in \mathbb{N}$*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \langle u_{n+h}, u_n \rangle = 0$$

then

$$\lim_{N \rightarrow \infty} \left\| \frac{1}{N} \sum_{n=1}^N u_n \right\| = 0.$$

Proof. Let $\varepsilon > 0$ and $H \in \mathbb{N}$. We have

$$\lim_{N \rightarrow \infty} \left\| \frac{1}{N} \sum_{n=1}^N u_n - \frac{1}{N} \frac{1}{H} \sum_{n=1}^N \sum_{h=0}^{H-1} u_{n+h} \right\| = 0 \quad (3.6)$$

(3.6) holds because the difference between first summation and the second sum in (3.6) is at most

$$\frac{1}{N} \left(\left\| \sum_{n=1}^{H-1} u_n \right\| + \left\| \sum_{n=N+1}^{N+H-1} u_n \right\| \right) = \frac{1}{N} O(1)$$

due to the fact that the sequence $(u_i)_{i=1}^\infty$ is bounded. We also have

$$\begin{aligned} \limsup_{N \rightarrow \infty} \left\| \frac{1}{N} \frac{1}{H} \sum_{n=1}^N \sum_{h=0}^{H-1} u_{n+h} \right\|^2 &\leq \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \left\| \frac{1}{H} \sum_{h=0}^{H-1} u_{n+h} \right\|^2 \\ &= \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \frac{1}{H^2} \sum_{h_1, h_2=0}^{H-1} \langle u_{n+h_1}, u_{n+h_2} \rangle \\ &\leq \frac{1}{H} \sup_{n \in \mathbb{N}} \|u_n\|^2. \end{aligned}$$

Taking the limit as $H \rightarrow \infty$ gives us

$$\lim_{H \rightarrow \infty} \limsup_{N \rightarrow \infty} \left\| \frac{1}{N} \frac{1}{H} \sum_{n=1}^N \sum_{h=0}^{H-1} u_{n+h} \right\|^2 = 0$$

which along with (3.6) implies that

$$\lim_{N \rightarrow \infty} \left\| \frac{1}{N} \sum_{n=1}^N u_n \right\| = 0$$

as desired. $\square$

This result implies Theorem 2.27. Take $\mathcal{H} = \mathbb{C}$ and for any equidistributed sequence $(x_n)_{n=1}^\infty$ take $u_n = e(x_n)$ in Theorem 3.11. The assumption in Theorem 2.27 can be equivalently stated as: for all $m \in \mathbb{Z} \setminus \{0\}$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \langle (u_{n+h})^m, (u_n)^m \rangle = 0.$$

With Theorem 3.11, this implies

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N (u_n)^m = 0$$

and by Weyl's criterion, we are done.

One immediate corollary of Theorem 2.27 is that $\mathbb{N}$ is a van der Corput set.

3.2.2 Van der Corput sets are sets of recurrence

In this subsection, we seek to identify some properties of van der Corput sets, in order to better understand their nature.

Lemma 3.12 *Van der Corput sets are always infinite.*

Proof. We will prove the contrapositive. Let us start with the simplest case: take the set $H = \{1\}$. In this case, it is not too hard to see that, for some sequence $(u_n)_{n=1}^\infty$ that is equidistributed, if we now take for $n \in \mathbb{N}$,

$$\begin{cases} x_{2n-1} = 0 \\ x_{2n} = u_n \end{cases}$$

then $(x_{n+1} - x_n)_{n=1}^\infty$ is equidistributed, as it alternates between u_n and $-u_n$, but $(x_n)_{n=1}^\infty$ is not (having half its terms sent to 0). This generalises to any singleton $H = \{k\}$ by taking a sequence that alternates between taking k 0s and k copies of u_n rather than just one. The question now becomes how we can generalise to arbitrary finite sets.

The first idea that is appealing is to use a similar setup to the previous case, where we have a nonzero density of values equal to 0. However, the difficulty we now face is ensuring the equidistribution of the sequences $x_{n+h} - x_n$. For this we use a potent idea with probabilistic flavour - randomness.

Let us first ensure the non-equidistribution of x_n . Take $H \subseteq \mathbb{N}$ a finite set and take $a > \max(H)$. Setting $x_a = x_{2a} = x_{3a} = \dots = 0$ ensures that the sequence x_n is not equidistributed.

Now, for all $i \neq 0 \pmod a$, we take x_i as i.i.d uniform $[0, 1]$ -random variables. Take $h \in H$. We have that

$$x_{n+h} - x_n = \begin{cases} x_{n+h} & \text{if } n = 0 \pmod a \\ -x_n & \text{if } n = -h \pmod a \\ x_{n+h} - x_n & \text{otherwise.} \end{cases}$$

The first thing to note is that, if we split the third case into subcases for all $a - 2$ other values that n can be modulo a , we have a cases in total (one for each modulo class). In all cases $x_{n+h} - x_n$ is a i.i.d uniform $[0, 1]$ -random variable. Note that the a cases are not independent between each other, but within each case, the terms are independent. If each individual case, the sequences are equidistributed, then the sequence of $x_{n+h} - x_n$ as a whole will be too. It therefore suffices to observe that a sequence of i.i.d uniform $[0, 1]$ -random variables will be equidistributed with probability 1. This is just an application of the law of large numbers, however, meaning that almost every sequence thus created will work. We leave the probabilistic details to the reader. $\square$

Indeed, this notion that equidistribution almost always occurs for randomly sampled sequences is quite useful for ensuring equidistribution without needing an explicit construction. Kamae and Mendès France[8] noted that the argument detailed in Lemma 3.12 generalised nicely for some infinite sets.

Lemma 3.13 *Consider a set $A \subseteq \mathbb{N}$ with positive upper density. Let $H \in \mathbb{N}$ be a set such that $(A - A) \cap H = \emptyset$. Then H is not a van der Corput set.*

Proof. The same argument as in Lemma 3.12 holds, except we now set $x_n = 0$ on A and i.i.d uniform $[0, 1]$ -random otherwise. Fixing h , it is impossible that two terms with difference h are both in A and indeed all cases we get can be shown to be equidistributed in a manner similar to the proof of Lemma 3.12. $\square$

One immediate corollary is that any van der Corput set is a set of recurrence, as in the proof of Lemma 3.13, any set H that is not a set of recurrence is not a van der Corput set, so the contrapositive is that van der Corput sets are a subset of sets of recurrence.

3.2.3 Ruzsa, Kamae and Mendès France's characterization of van der Corput sets.

We shift our focus towards finding more specific sufficient conditions for sets that are van der Corput. It follows from the papers of Ruzsa[22] and Kamae and Mendès France[8] that the following result holds, which will be the basis of this section.

Theorem 3.14 *Let $H \subseteq \mathbb{N}$. The following statements are equivalent:*

- (i) *H is a van der Corput set.*
- (ii) *If μ is a positive measure on the torus $\mathbb{T}$ such that, for all $h \in H$, $\hat{\mu}(h) = 0$, then $\mu(\{0\}) = 0$.*

The direction (ii) $\implies$ (i) was proven by Kamae and Mendès France in 1978, whilst Ruzsa was the first to note that both statements were equivalent in 1984.

The reader should recall that exponential polynomials, trigonometric polynomials and their spectrum were defined in Section 2.3, and that trigonometric and exponential polynomials always have a finite spectrum.

We will begin with discussing the paper of Kamae and Mendès France. The key to their proof lies in harmonic analysis. We begin with the following definition of *positive-definite* sequences:

Definition 3.15 (Positive definite) *We call a function $f: \mathbb{R} \rightarrow \mathbb{C}$ positive definite if, for any finite sequence of complex numbers $(y_i)_{i=1}^n$ we have*

$$\sum_{k,l=1}^n f(k-l)y_k\overline{y_l} \geq 0.$$

Following this definition, we call a sequence of complex numbers $(x_m)_{m=-\infty}^{\infty}$ positive definite if there exists a positive definite function f such that $f(m) = x_m$.

Note that if $f: \mathbb{R} \rightarrow \mathbb{C}$ is a positive definite function then $f(0) \geq f(m)$ for all $m \in \mathbb{Z}$. It is of interest to find examples of a positive definite sequence, since it is not even immediate that one exists - the fact that $\sum_{k,\ell=1}^n x_{k-\ell} y_k \overline{y_\ell}$ should be real with no imaginary part is not at all trivial, let alone positive. We have a clue in the expression itself - $x_{k-\ell} y_k \overline{y_\ell}$ - that motivates choosing $x_{k-\ell} = e(k-\ell)$ so that we can make a second complex conjugate appear. Indeed, we have

$$\begin{aligned} \sum_{k,\ell=1}^n e(k-\ell) y_k \overline{y_\ell} &= \sum_{k,\ell=1}^n e^{2\pi i k} y_k \overline{e^{2\pi i \ell} y_\ell} \\ &= \left(\sum_{k=1}^n e^{2\pi i k} y_k \right) \overline{\left(\sum_{\ell=1}^n e^{2\pi i \ell} y_\ell \right)} \end{aligned}$$

where it is easy to see that the two expressions are equal and hence we obtain the product of a complex number with its conjugate, which is always nonnegative.

We state the following result, without proof. Recall that $\hat{f}$ and $\hat{\mu}$ refer to the Fourier transforms of a function or a measure (Definition 2.20, Definition 2.21).

Lemma 3.16 (Bochner-Herglotz's theorem) *Given a nonzero positive definite sequence of complex numbers $(y_m)_{m=-\infty}^{\infty}$ where $|y_0| \leq 1$, there exists a positive Borel measure μ on the torus such that $\hat{\mu}(m) = y_m$.*

A proof of Bochner-Herglotz's theorem can be found in Helson's book[23]. Helson notes that this works in both directions: given a positive Borel measure μ on the torus, $(\hat{\mu}(m))_{m=-\infty}^{\infty}$ is a positive-definite sequence. Typically in literature, one calls μ the *spectral measure* associated to the sequence $(\hat{\mu}(m))_{m=-\infty}^{\infty}$. We now proceed with a new definition on complex sequences.

The following definition is taken from Betrandias[24], and is a discrete version of an earlier definition by Bass[25].

Definition 3.17 (Pseudorandom sequence) *Given a sequence of complex numbers $(x_m)_{m=-\infty}^{\infty}$ we define the sequence $(\gamma(k))_{k=1}^{\infty}: \mathbb{Z} \rightarrow \mathbb{C}$ as*

$$\gamma(k) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} x_{n+k} \overline{x_n} \quad (3.7)$$

if the limit exists for all $k \in \mathbb{Z}$. If furthermore we have that

$$\lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{k=-n}^n |\gamma(k)|^2 = 0$$

then we call the sequence $(x_m)_{m=-\infty}^{\infty}$ pseudorandom.

We have the following immediate property of pseudorandom sequences.

Lemma 3.18 *Given a pseudorandom sequence $(x_m)_{m=-\infty}^{\infty}$ of complex numbers, we have that the sequence $(\gamma(k))_{k=-\infty}^{\infty}$ as defined in (3.7) Definition 3.17 is positive-definite.*

This is a straightforward application of the definition of positive-definiteness; this result is the first theorem in a paper of Bass[25] on pseudorandom sequences and the proof can be found there. It immediately follows that we can now apply Bochner-Herglotz to find a positive Borel measure μ from the positive-definite sequence $(\gamma(k))_{k=-\infty}^{\infty}$. Furthermore, via a result of Wiener[26] in his paper on harmonic analysis (Chapter 6), we know that the measure μ is continuous and does not have point masses. This will prove to be very useful for showing equidistribution.

Definition 3.19 *Let $H \subseteq \mathbb{N}$. We denote by $P(H)$ the set of all exponential polynomials $f \in L_1(\mathbb{T})$ such that*

- (i) $f(0) = 1$;
- (ii) $|f(x)| \leq 1$;
- (iii) *The spectrum of f is a subset of H .*

We denote by $P^(H)$ the closure of $P(H)$ under pointwise convergence (which is still a subset of $L_1(\mathbb{R})$), and by $P_\delta^*(H)$ the subset*

$$\{f \in P^*(H) : \Re(f(x)) \geq -\delta \forall x \in \mathbb{T}\}.$$

We now introduce an auxiliary result from the paper of Kamae and Mendès France[8].

Theorem 3.20 *Let $(\hat{\mu}(m))_{m=-\infty}^{\infty}$ be a positive definite sequence with spectral measure μ such that $|\mu(\{0\})| \leq 1$. Let $H \subseteq \mathbb{N}$ be a subset of the positive integers where*

- (i) *for some $\delta \geq 0$, $P_\delta^*(H) \neq \emptyset$,*
- (ii) $\forall n \in H, \hat{\mu}(n) = 0$.

Then $\mu(\{0\}) \leq \delta$.

Proof. Consider a subset $H \subseteq \mathbb{N}$, a sequence $(\hat{\mu}(m))_{m=-\infty}^{\infty}$ as described in the statement of Theorem 3.20 and its spectral measure μ .

Firstly, we have the following observation: Take any exponential polynomial $f \in P(H)$. We then have the property

$$f(x) = \sum_{n \in \mathbb{N}} \hat{f}(n) e(-2\pi n x) = \sum_{n \in H} \hat{f}(n) e(-2\pi n x)$$

and furthermore

$$\int_T f(x) \mu(dx) = \sum_{n \in H} \hat{f}(n) \hat{\mu}(n) = 0 \quad (3.8)$$

with the latter equality being a consequence of our initial assumption about $\hat{\mu}$. This property extends to $P^*(H)$ by pointwise convergence.

Now consider $P_\delta^*(H)$ and pick some element $f \in P_\delta^*(H)$. In particular

$$\Re(f(x)) \geq -\delta \implies \int_T \Re(f(x)) \mu(dx) \geq -\delta.$$

Indeed, we can improve this to

$$\int_T \Re(f(x)) \mu(dx) \geq \mu(\{0\}) - \delta.$$

as the mass at 0, if it is nonzero, does not contribute anything to the integral. This is of course true for any singleton and not just 0, but for reasons we will see we are only interested in this specific bound.

Recalling (3.8), this implies that $\mu(\{0\}) \leq \delta$. $\square$

It turns out we can show that sets H as described in Theorem 3.20 are van der Corput. We first introduce a special class of pseudorandom sequences noted by Bertrandias[24] in his 1964 paper.

Lemma 3.21 *Let $(x_i)_{i=1}^\infty$ be a sequence of reals with the property that $x_n = o(\sqrt{n})$ as $n \rightarrow \infty$ and*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N |x_n|^2 \leq 1.$$

Then, the limit

$$\hat{\mu}(m) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N x_n \overline{x_{n+m}} \quad (3.9)$$

exists and by Lemma 3.18 the sequence $(\hat{\mu}(m))_{m=-\infty}^\infty$ is positive definite. Furthermore, if μ is the spectral measure induced by $\hat{\mu}$ via Bochner-Herglotz Theorem, then

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \left| \sum_{n=1}^N x_n \right| \leq \mu(\{0\}). \quad (3.10)$$

We will not prove this result here; the proof can be found in the first section of Bertrandias'[24] paper.

Theorem 3.22 *If $H \subseteq \mathbb{N}$ is a set such that $P_\delta^*(H) \neq \emptyset$ for all $\delta > 0$, then H is a van der Corput set.*

Proof. Let $(x_i)_{i=1}^\infty$ be any sequence such that for all $h \in H$, $(x_{i+h} - x_i)_{i=1}^\infty$ is equidistributed modulo 1. Now, if

$$y_i := e(x_i)$$

then $(y_i)_{i=1}^\infty$ is a sequence satisfying the conditions in the statement of Lemma 3.21 that gives in turn a positive-definite sequence from (3.9) that we call $(\hat{\mu}(m))_{m=-\infty}^\infty$.

By Weyl's criterion, we know that $\forall h \in H$, $\ell \in \mathbb{N}$ (taking here $\ell = 1$), we have

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e(\ell(x_{n+h} - x_n)) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e(x_{n+h} - x_n) = 0$$

and the same is therefore true for

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e(x_n - x_{n+h}) = 0$$

since this is just the sum of the complex conjugates and hence the limit is the complex conjugate of 0, hence 0. However, since

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e(x_n - x_{n+h}) &= \\ \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e(x_n) \overline{e(x_{n+h})} &= \\ \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N y_n \overline{y_{n+h}} &= \mu(h) \end{aligned}$$

with the last equality following from our prior usage of Lemma 3.21 then it follows that for all $h \in H$, $\mu(h) = 0$. Along with our initial assumption that $P_\delta^*(H) \neq \emptyset$ for all $\delta > 0$, we can now apply Theorem 3.20 to obtain $\mu(\{0\}) = 0$, implying by (3.10) from Lemma 3.21 that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \left| \sum_{n=1}^N y_n \right| = \lim_{N \rightarrow \infty} \frac{1}{N} \left| \sum_{n=1}^N e(x_n) \right| = 0.$$

But this immediately implies, via Weyl's criterion again, that $(x_i)_{i=1}^\infty$ is also distributed mod 1. $\square$

If we assume, given a set $H \subseteq \mathbb{N}$, that for all measures μ satisfying $\hat{\mu}(h) = 0$ for all $h \in H$ we have $\mu(\{0\}) = 0$, then the proof is much shortened and one direction in Theorem 3.14 follows immediately. Let us proceed to the other direction: if we have a Borel measure μ on the torus whose spectrum does not contain any element of a van der Corput set H , then $\mu(\{0\}) = 0$.

Ruzsa[22] would be the first to prove this. His 1981 paper established the following definition, which he mentions was originally suggested by Mendès France.

Definition 3.23 (Correlative set) *We say a $H \subseteq \mathbb{N}$ is a correlative set if it has the following property: for any sequence $(x_i)_{i=1}^{\infty}$ of complex numbers, the two assumptions*

- (i) $\sum_{n=1}^N |x_n|^2 = O(N),$
- (ii) $\sum_{n=1}^N \overline{x_n} x_{n+h} = o(N) \forall h \in H$

imply that

$$\sum_{n=1}^N |x_n| = o(N).$$

It turns out this definition is equivalent to being a van der Corput set, along with a couple of others.

Theorem 3.24 *The following four definitions are equivalent for $H \subseteq \mathbb{N}$:*

- (i) H is a correlative set.
- (ii) H is a van der Corput set.
- (iii) *For every $\varepsilon > 0$ there is a trigonometric polynomial $f(x) = \sum_{h \in H} a_h \cos(2\pi hx)$ (note that we write it in the trigonometric rather than the exponential form) such that $f(0) = 1, f(x) > -\varepsilon$ for all x .*
- (iv) *Every measure μ on $\mathbb{T}$ such that*

$$\int_{\mathbb{T}} \cos(2\pi hx) d\mu(x) = 0$$

holds for all $h \in H$, has no point mass at 0 (i.e., $\mu(\{0\}) = 0$).

In order to prove that (i), (ii), (iii) and (iv) are equivalent, Ruzsa defined a notion of "distance" for each of the four definitions that would be equal to 0 when a set satisfied the definition.

Firstly, (i) corresponds to van der Corput sets. We begin by giving a standard definition that relates closely to equidistribution. In what follows, the sequences $(x_i)_{i=1}^{\infty}$ are assumed to be taken modulo 1.

Definition 3.25 (Discrepancy) *The discrepancy of the first n terms of a sequence $(x_i)_{i=1}^{\infty}$ is defined as*

$$D_n((x_i)_{i=1}^{\infty}) = \sup_{0 \leq a \leq b \leq 1} \left| \frac{|\{x_1, \dots, x_n\} \cap [a, b]|}{n} - b - a \right|$$

It is easy to see that equidistribution is equivalent to $\lim_{n \rightarrow \infty} D_n = 0$.

For $H \subseteq \mathbb{N}$ let $\mathcal{X}(H)$ be the family of sequences such that, for any sequence $(x_i)_{i=1}^\infty \in \mathcal{X}(H)$ and any $h \in H$, $(x_{n+h} - x_n)_{n=1}^\infty$ is equidistributed. If all these sequences are equidistributed as well, then H is of course van der Corput.

We therefore define

$$\alpha(H) := \sup_{(x_i)_{i=1}^\infty \in \mathcal{X}(H)} \limsup D_n((x_i)_{i=1}^\infty)$$

and it is clear that $\alpha = 0$ is equivalent to H having the van der Corput property.

Similarly, for (ii) in Theorem 3.24, let $H \subseteq \mathbb{N}$ and let $\mathcal{Y}(H)$ be the family of sequences such that, for any sequence $(y_i)_{i=1}^\infty \in \mathcal{Y}(H)$ we have both

$$\sum_{n=1}^N \overline{y_n} y_{n+h} = o(N) \quad \forall h \in H \quad (3.11)$$

and

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N |y_n|^2 \leq 1. \quad (3.12)$$

Note that (3.11) and (3.12) are similar to (ii) and (i) respectively in Definition 3.23. We then define

$$\beta(H) := \sup_{(y_i)_{i=1}^\infty \in \mathcal{Y}(H)} \limsup_{N \rightarrow \infty} \left| \frac{1}{N} \sum_{n=1}^N y_n \right| \quad (3.13)$$

and it is clear that $\beta = 0$ is equivalent to H being correlative. We also define $\beta^*(H)$ as the value in (3.13) with the stronger additional constraint that $|y_n| \leq 1 \quad \forall n \in \mathbb{N}$.

For (iii) in Theorem 3.24, let $\mathcal{F}(H)$ be the set of trigonometric polynomials $f(x) = a_0 + \sum_{h \in H} a_h \cos(2\pi hx)$ that are nonnegative and satisfy $f(0) = 1$, and we define

$$\gamma(H) = \inf_{f \in \mathcal{F}(H)} a_0$$

and, once again, $\gamma = 0$ corresponds to the definition stated in the theorem. Last but not least, for (iv) in Theorem 3.24 take the set $\Omega(H)$ of measures μ on $\mathbb{T}$ satisfying

$$\int_{\mathbb{T}} \cos(2\pi hx) d\mu(x) = 0 \quad \forall h \in H$$

and define

$$\lambda(H) = \sup_{\mu \in \Omega(H)} \mu(\{0\})$$

the largest point mass at 0, with $\mu = 0$ if and only if (iv) is satisfied. We therefore want to show

$$\alpha(H) = 0 \iff \beta(H) = 0 \iff \beta^*(H) \iff \gamma(H) = 0 \iff \lambda(H) = 0.$$

Note that we did not explain explicitly why all of $\alpha, \beta, \beta^*, \gamma$ and λ always exist but it is easy to verify that the sets $\mathcal{X}(H), \mathcal{Y}(H), \mathcal{F}(H)$ and $\Omega(H)$ are always nonempty for any $H \subseteq \mathbb{N}$. We will present here a few of the relationships we can deduce between $\alpha, \beta, \beta^*, \gamma$ and λ , and state the others without proof. This proof sketch contains several auxiliary lemmas (3.26, 3.27, 3.28 and 3.29) along with some of their proofs.

Proof. (Proof sketch of Theorem 3.24) We state a series of bounds on our different "indicators" that together imply they are equal to 0 at the same time.

Lemma 3.26 *For all $H \subseteq \mathbb{N}$, we have*

$$\beta^2(H) \leq \gamma(H).$$

Proof. (Lemma 3.26) Returning to the definition of γ , take $f(x) \in \mathcal{F}(H)$. By the Fejér-Riesz theorem, there exists another trigonometric polynomial $g(x)$ such that

$$f(x) = |g(x)|^2$$

and since g has finite spectrum, take $M \in \mathbb{N}$ such that $\widehat{g}(m) = 0$ if $|m| > M$. We can further assume that $g(0) = 1$ as the Fourier coefficients of g must sum to 1, the squares of the coefficients must sum to $\widehat{f}(0)$, and more generally

$$2 \sum_{n \in \mathbb{N}} \widehat{g}(n+k) \overline{\widehat{g}(n)} = \widehat{f}(k). \quad \square$$

Now take $(y_m)_{m=-\infty}^{\infty}$ a sequence satisfying the conditions in (3.11) and (3.12), letting $y_m = 0$ for $m \leq 0$. We want to find useful bounds on the sum $\sum_{n=1}^k y_n$. We notice that if we define Y_k to be the finite sum of the first k terms and

$$Z_k = \sum_{n=1}^k \sum_{m=-M}^M \widehat{g}(m) y_{n+m}$$

then we note already that

$$Z_k = Y_k + O(\sqrt{k}) \quad (3.14)$$

since, as mentioned, $\sum_{m=-M}^M \widehat{g}(m) = 1$ and so

$$\begin{aligned} Z_k - Y_k &= \sum_{n=1}^k y_n - \sum_{n=1}^k \sum_{m=-M}^M \widehat{g}(m) y_{n+m} \\ &= \sum_{n=1}^k y_n - \sum_{n=1}^k \left(y_n \left(1 - \sum_{m=\max(-M, n-k)}^{\min(M, n)} \widehat{g}(m) \right) \right) \\ &= \sum_{n=1}^k \left(y_n \left(- \sum_{m=\max(-M, n-k)}^{\min(M, n)} \widehat{g}(m) \right) \right) \end{aligned}$$

Along with the assumptions in (3.11) and (3.12) (which notably imply that $y_n = O(\sqrt{n})$), this shows (3.14). Applying AM-QM, we further have

$$\frac{1}{k} Z(k)^2 \leq \sum_{\ell, n, m \in \mathbb{N}} \widehat{g}(n) \overline{\widehat{g}(m)} y_{\ell+n} y_{\ell+m}.$$

If we now look at all the pairs n, m where $|n - m| = j$ then by our assumptions on $g(x)$ and $(y_m)_{m=-\infty}^{\infty}$,

$$\frac{1}{k} Z(k)^2 \leq (\widehat{f}(0) + o(1))k + O\left(\sum_{|i-M| \leq M} |y_i|^2\right).$$

We can indeed show that the last term here cannot be large too often (the interested reader is invited to check Ruzsa's paper for the proof) and it follows that for some $k' \in (k, k + \sqrt[3]{k})$,

$$Y_{k'} \leq (\widehat{f}(0) + o(1))k.$$

We can also calculate that, since $k' \in (k, k + \sqrt[3]{k})$,

$$|Y_{k'} - Y_k| \leq k^{\frac{5}{6}}$$

as the difference is just additional terms $\sum_{n=k}^{k'} Y_n$, and we can use

$$\left| \sum_{n=k}^{k'} Y_n \right| \leq \sum_{n=k}^{k'} |Y_n|$$

and applying Cauchy-Schwarz to the latter sum, along with again the fact that $y_n = O(\sqrt{n})$, gives the desired bound. This implies that

$$\limsup \left| \frac{1}{k} Y_k \right| \leq \sqrt{\widehat{f}(0)} \quad (3.15)$$

which is exactly equivalent to $\beta^2 \leq \gamma$, since $\widehat{f}(0) = a_0$ and (3.15) holds for all $f \in \mathcal{F}(H)$.

Lemma 3.27 *For all $H \subseteq N$, such that both $\lambda(H)$ and $\gamma(H)$ exist, we have*

$$\lambda = \gamma.$$

We will not prove this specific relationship here as it is clear that some relationship exists (unlike the other inequalities presented so far, which connect definitions that are quite significantly different) and show this exact lemma is a matter of careful functional analysis which the curious reader can find in Ruzsa's paper.

Lemma 3.28 *For all $H \subseteq N$, such that all of $\lambda(H)$ and $\beta^*(H)$ and $\alpha(H)$ exist, we have*

$$\lambda \leq \beta^* \leq 2\alpha.$$

We will not present the proof of this lemma either but it uses the idea of randomly sampling sequences and showing w.h.p the desired inequalities hold.

Lemma 3.29 *For all $H \subseteq N$ such that both $\alpha(H)$ and $\beta^*(H)$ exist we have*

$$\alpha \leq \beta^* \log \left(\frac{2}{\beta^*} \right).$$

Proof. (Lemma 3.29) This follows from a result of Erdős and Turán[27], which allows us to bound the discrepancy. The exact statement of the result is that if $(x_i)_{i=1}^\infty$ is a sequence of reals, then defining

$$\theta_t(k) := \frac{1}{k} \left| \sum_{n=1}^k e(tx_n) \right|$$

one has the inequality, for any k ,

$$D_n((x_i)_{i=1}^\infty) \leq \frac{1}{k} + \sum_{i=1}^k \frac{1}{t} \theta_t(i).$$

If we now take any sequence $(x_i)_{i=1}^\infty \in \mathcal{X}(H)$, then fixing some $t \in \mathbb{N}$ and considering $y_i = e(tx_i)$, we already know by the definition of β^* that taking any $h \in H$, we have

$$\sum_{i=1}^k \overline{y_i} y_{i+h} = \sum_{i=1}^k e(k(x_{i+h} - x_i)) = o(x).$$

It follows that, asymptotically,

$$\theta_t(k) = \frac{1}{k} \left| \sum_{i=1}^k e(tx_i) \right| = \frac{1}{k} \left| \sum_{i=1}^k y_i \right| \leq \beta^* + o(1)$$

for any choice of $t \in \mathbb{N}$. Applying the result of Erdős and Turán, we conclude that for any choice of $m \in \mathbb{N}$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} D_n((x_i)_{i=1}^\infty) \leq \frac{1}{m} + \beta^* \log(2m) \leq \beta^* \log\left(\frac{2}{\beta^*}\right)$$

where the last inequality follows by taking $m \geq \frac{1}{\beta^*}$. $\square$

As a result, it is now clear that by the bounds we have just proven, if any of $\alpha, \beta, \beta^*, \gamma, \lambda$ equals 0, then so do the rest. This concludes the proof sketch of Theorem 3.24. $\square$

The most important result for us is Lemma 3.28, which implies in particular that for a van der Corput set H , one has that every measure on the torus satisfying

$$\int_{\mathbb{T}} \cos(2\pi hx) d\mu(x) = 0 \quad (3.16)$$

for all $h \in H$ has $\mu(\{0\}) = 0$. In particular, if we now have a measure μ such that for all $h \in H$, $\hat{\mu}(h) = 0$ then, employing Euler's formula, (3.16) holds and so $\mu(\{0\}) = 0$. This completes the proof of Theorem 3.14.

3.3 Sets of recurrence that are not van der Corput

It is an instinctive follow-up to now ask oneself whether there is an equivalence between sets of recurrence and van der Corput sets. We already proved that all van der Corput sets are sets of recurrence (Lemma 3.13), and all our examples of sets of recurrence happen to be van der Corput as well. The answer is in the negative, and was first shown by Bourgain[9] in 1987. We will showcase Mountakis'[10] construction of such a set, which is inspired by Bourgain's original work. As a matter of fact, Mountakis' proof gives a set of *strong recurrence* that is not van der Corput. We will define the notion of strong recurrence in Definition 3.43.

3.3.1 Generalisations and more Fourier analysis

We need some preliminaries to prove Mountakis' result using the more general definitions. In particular, we will introduce variants of the notions of sets of recurrence and van der Corput sets.

Definition 3.30 (Sets of ε -recurrence) Let $\varepsilon > 0$ and consider a set $R \subseteq \mathbb{N}$. We say that R is a set of ε -recurrence if there exists some $n \in \mathbb{N}$ such that for all sets $E \subseteq [n]$ with $|E| > \varepsilon n$,

$$(E - E) \cap R \neq \emptyset.$$

This is a weakening of Definition 3.3. As shown before, a set $H \subseteq \mathbb{N}$ is an intersective set if and only if it is a set of recurrence. Therefore, a set of recurrence is a set of ε -recurrence for all ε . We will give an example of a set of ε -recurrence that is not a set of recurrence on the next page.

If Definition 3.30 holds for some n , then it holds for all larger numbers, as we show in the following proposition.

Proposition 3.31 *Let $\varepsilon > 0$ in $\mathbb{R}$. If $R \subseteq \mathbb{N}$ is a set of ε -recurrence and $n \in \mathbb{N}$ is any positive integer for which the properties described in Definition 3.30 hold, then for every $\delta > 0$ there exists $N \in \mathbb{N}$ such that for all $M \geq N$ and all sets $E \subseteq [M]$ with $|E| > (\varepsilon + \delta)M$,*

$$(E - E) \cap R \neq \emptyset.$$

Before we proceed to the proof, let us ask ourselves why a δ is necessary in the statement of Proposition 3.31. Essentially, what we want to show here is that if a set E has density ε as a subset of $[M]$, then there exists k such that $E \cap \{k+1, k+1, \dots, k+n\} \geq \varepsilon n$ - in other words, there is a subsection over which the density is at least ε . Then by the trivial observation that $(E - E) = ((E - \ell n) - (E - \ell n))$ (in other words, the difference set of a set is obviously shift-invariant) we can essentially reduce E to a set satisfying the conditions in Definition 3.30, allowing us to indeed conclude that $(E - E) \cap R \neq \emptyset$.

Unfortunately, this is not true in principle - consider the odd numbers from 1 to 99 for example; they have density $\frac{45}{99}$ on this set, but over any consecutive four integers, they have density $\frac{1}{2}$ which is lower. What is clear is that for any multiple kn of n , if E has density at least ε as a subset of $[kn]$ then we can do what we describe - just divide $[kn]$ into k groups of n consecutive integers, and by the pigeonhole principle, one of these groups (say $\{\ell n + 1, \ell n + 2, \dots, (\ell + 1)n\}$) has at least εn terms in E as desired. So if M is not a multiple of n , it therefore suffices to augment ε slightly to ensure that there is some subsection of sufficient density.

Proof. (Proposition 3.31) Suppose R is a set of ε -recurrence. Then, by definition there exist n such that for any subset E of $[n]$ with density greater than ε , $(E - E) \cap R \neq \emptyset$. For any $M \geq \frac{n}{\delta}$, and any subset E' of $[M]$ with density greater than $\varepsilon + \delta$, one also has $(E' - E') \cap R \neq \emptyset$. Notice if M is a multiple of n it is true by pigeonhole; if $M \equiv r \neq 0 \pmod{n}$ then look at the last $n + r$ numbers. Either they are all in E' (in which case we obviously have $(E' - E') \cap R \neq \emptyset$) or they are not and the density of E' as a subset of

3.3. Sets of recurrence that are not van der Corput

$[M - n - r]$ is at least ε , because $\frac{r}{M} < \frac{n}{M} \leq \delta$, meaning

$$\begin{aligned} \frac{|E' \cap [M - n - r]|}{M - n - r} &> \frac{|E' \cap [M - n - r]|}{M} \\ &= \frac{|E' \cap [M]|}{M} - \frac{|E' \cap \{M - n - r, \dots, M\}|}{M} \\ &> (\varepsilon + \delta) - \delta = \varepsilon. \end{aligned}$$

This allows us to conclude by pigeonhole with $E' \cap [M - n - r]$, as $M - n - r$ is a multiple of n . $\square$

Let us introduce the notion of ε -van der Corput sets.

Definition 3.32 (Sets of ε -vdC) Let $\varepsilon > 0$ and let $H \subseteq \mathbb{N}$. We say that H is an ε -vdC set if, for every measure μ on $\mathbb{T}$ such that

$$\widehat{\mu}(h) = 0$$

holds for all $h \in H$, one has $\mu(\{0\}) \leq \varepsilon$.

Notice that van der Corput sets are ε -vdC for every ε , by Theorem 3.14.

We have not yet commented on whether van der Corput sets are a strict subset of ε -vdC sets, or the analogous statement with sets of recurrence and ε -recurrence. Both of these are true and we will see this over the course of this proof, but we can already comment on the latter inclusion: it is easy to see that, for $N \geq 2$, $E = [N - 1]$ is a set of $\frac{1}{N-1}$ -recurrence as any subset of $[N]$ with at least $\frac{1}{N-1} \cdot N > 1$ elements has two elements and therefore a difference in E . More specifically, we can find *finite* sets of ε -recurrence for any fixed $\varepsilon > 0$, which are obviously not sets of recurrence.

Before we proceed to the the proof of Mountakis' result, we will need a couple more tools from Fourier analysis.

Definition 3.33 (Convolution of two measures on $\mathbb{T}$) If μ_1, μ_2 are non-negative finite measures on $\mathbb{T}$, their convolution $\mu_1 * \mu_2$ is defined as the unique non-negative finite measure on $\mathbb{T}$ satisfying

$$\int_{\mathbb{T}} f(t) d(\mu_1 * \mu_2)(x) = \int_{\mathbb{T}^2} f(t_1 + t_2) d\mu_1(t_1) d\mu_2(t_2)$$

for all continuous functions f on $\mathbb{T}$ satisfying $f(0) = f(1)$.

It is known that such a measure always exists and is unique, as Katznelson[28] shows in his book on page 4. The Fourier transform of the convolution is the product of the transforms; in other words:

$$\widehat{\mu_1 * \mu_2} = \widehat{\mu_1} \widehat{\mu_2}.$$

Definition 3.34 (Fejér kernel) *The Fejér kernel is the sequence of trigonometric polynomials $(F_i)_{i=1}^\infty$ where $F_n: \mathbb{T} \rightarrow \mathbb{C}$ defined by*

$$F_n(t) := \sum_{k=-n+1}^{n+1} \left(1 - \frac{|k|}{n}\right) e(kt) \quad (3.17)$$

One can show that

$$F_n(t) = \frac{1}{n} \left(\frac{\sin(\pi n t)}{\sin(\pi t)} \right)^2 \quad (3.18)$$

by rewriting $e(kt)$ as a sum of imaginary and real trigonometric parts as done before, and cancelling and simplifying the expression.

F_n is clearly positive; furthermore, it attains its maximum at 0 where it is equal to n . We also have

$$F_n(t)F_m(nt) = F_{nm}(t)$$

(note the nt rather than t as the argument of F_m) which is not obvious from the definition but follows naturally from the property mentioned just after.

Proposition 3.35 *The Fourier coefficients of F_M are given by:*

$$\hat{F}_M(m) = \begin{cases} 0 & \text{if } |m| \geq M \\ 1 - \frac{|m|}{M} & \text{if } |m| \leq M \end{cases}.$$

This is immediate from (2.1).

One can find a detailed exposition of the Fejér kernel and all these properties and more in Katznelson's book[28] stating on page 12.

We need a couple of other properties of the Fejér kernel. Firstly, positive trigonometric polynomials can be bounded by their convolution with a Fejér kernel.

Lemma 3.36 *Let R, L be positive integers and $f = \sum_{|m| \leq RL} e^{2\pi i m t}$ be a trigonometric polynomial, and assume furthermore that $f \geq 0$. Then $f \leq 4Rf * F_L$.*

Proof. We will lean into a useful property of Fejér kernels, which is that their Fourier transforms are also quite nice, as noted in Proposition 3.35. In particular, if we let

$$K := 2F_{2RL} - F_{RL} \quad (3.19)$$

then we have $\widehat{K}(m) = 1$ for $|m| \leq RL$. Using (3.19), it follows that

$$\widehat{f}(m) = \widehat{f * K}(m)$$

since the spectrum of f is a subset of the spectrum of K . In particular, $f = f * K$. By definition (and the positivity of the Fejér kernels) we know $K \leq 2F_{2RL}$. We can further bound $2F_{2RL}$ since $F_{2R}(Lt)F_L(t) = F_{2RL}$, and $F_{2R}(Lt) \leq 2R$. This gives

$$f = f * K \leq f * 2F_{2RL} \leq 4Rf * F_L$$

as desired. $\square$

Secondly, if a sequence of reals $(x_n)_{n=0}^\infty$ is nonnegative, decreasing and convex¹ and is equal to 0 somewhere then the trigonometric polynomial whose coefficients are taken from the sequence is nonnegative.

Lemma 3.37 *If a sequence of reals $(x_n)_{n=0}^\infty$ is nonnegative, decreasing and convex and satisfies $x_\ell = 0$ for some $\ell \in \mathbb{N}_0$, then, defining the trigonometric polynomial*

$$f(t) = \sum_{m \in \mathbb{Z}} x_{|m|} e(mt)$$

we have that $f(t) \geq 0$.

Proof. Note that the sequence is equal to 0 from some term x_ℓ onwards by our assumption and so f is again a trigonometric polynomial. We will proceed by induction on the first index ℓ where x_ℓ is equal to 0. If $\ell = 0$ then the sequence must be constantly equal to 0 so we are done. Assume now that the lemma is true up to some ℓ and consider a sequence which is equal to 0 from $x_{\ell+1}$ onwards. Define

$$y_n = x_n - x_\ell(\ell+1)\widehat{F_{\ell+1}}(n)$$

where we observe that $y_\ell, y_{\ell+1}, \dots = 0$ and for $n < \ell$,

$$\begin{aligned} x_\ell(\ell+1)\widehat{F_{\ell+1}}(n) &= x_\ell(\ell+1) \left(1 - \frac{n}{\ell+1}\right) \\ &\leq x_\ell(\ell+1-n) \\ &\leq x_n \end{aligned}$$

where the last inequality follows by convexity, namely that

$$x_\ell \leq \frac{1}{\ell+1-n}x_n + \frac{\ell-n}{\ell+1-n} \underbrace{x_{\ell+1}}_0$$

so we can calculate that $y_n \geq 0$. As $x_\ell(\ell+1)\widehat{F_{\ell+1}}$ is convex and decreasing, so is y_n . Additionally, $y_\ell = 0$ so by assumption if

$$g(t) = \sum_{m \in \mathbb{Z}} y_{|m|} e(mt)$$

¹in the sense that $x_{tn+(1-t)m} \leq tx_n + (1-t)x_m$, for $n < m \in \mathbb{N}$ and $t \in \{0, \frac{1}{m-n}, \dots, \frac{m-n-1}{m-n}, 1\}$

then $g(t)$ is nonnegative. However, we have

$$f(t) - g(t) = x_\ell(\ell + 1) \sum_{m \in \mathbb{Z}} \widehat{F_{\ell+1}} e(mt) = f(\ell)(\ell + 1)F_{\ell+1}$$

which is real and nonnegative by the properties of the Fejér kernel. This means that f , as the sum of two real and nonnegative functions, is itself a real and nonnegative function. $\square$

3.3.2 Mountakis' result

In this subsection, we proceed towards the construction of a set of recurrence that is not van der Corput. We begin with the following.

Theorem 3.38 *For every $0 < \delta, \varepsilon \leq \frac{1}{2}$ there is a finite set $R \subseteq \mathbb{N}$ which is a set of δ -recurrence but not an ε -vdC set.*

This is the main result showcased in this thesis, and the proof will last the entirety of this subsection, and contain several auxiliary results (Lemma 3.39, Lemma 3.41, Theorem 3.40, Lemma 27).

First note that due to the trivial fact that, for any $\delta_1 < \delta_2$, a set of δ_1 -recurrence is also a set of δ_2 -recurrence, it suffices to prove Theorem 3.38 for arbitrarily small δ . In particular, we will take $\delta = \frac{1}{j}$ for some $N \in \mathbb{N}$, and provide a corresponding set R_j .

We must first consider a lemma of a combinatorial nature that will be useful for the proof. Here, when we work modulo an integer N , we treat the equivalency classes like the integers $0, \dots, N-1$, and for $x \in \mathbb{Z}$, $x \bmod N$ is equal to the unique integer $r \in \{0, \dots, N-1\}$ for which there exists $k \in \mathbb{Z}$ such that $x = kN + r$. We will refer to N -uples as "points".

Lemma 3.39 *Let $\ell, M \in \mathbb{N}$ and consider any $N \in \mathbb{N}$ such that $N > 2M \log \ell$. Consider the space $[\mathbb{Z}/2M\mathbb{Z}]^N$ of N -uples modulo $2M$; denote an element of this space x as $(x_0, \dots, x_{N-1})$ with each of the coordinates in $\mathbb{Z}/2M\mathbb{Z}$. Equip $\mathbb{Z}/2M\mathbb{Z}$ with the metric $d(a, b) = \min(a - b \bmod 2M, b - a \bmod 2M)$. Then for any set of points $S \subseteq [\mathbb{Z}/2M\mathbb{Z}]^N$ with $|S| \geq \frac{1}{\ell}(2M)^N$ we have two points $x, y \in S$ and some integer $1 \leq s \leq N-1$ such that*

- (i) $x_i = y_i$ for $0 \leq i \leq s-1$;
- (ii) $x_s = 0, y_s = M$;
- (iii) $d(x_i, y_i) \leq 2$ for $s+1 \leq i \leq N-1$.

This lemma is essentially a quantification of the minimum number of points needed in this space to have two that are equal on the first s coordinates, are antipodal (specifically at 0 and M) at the $(s+1)$ th coordinate and then are "close" with respect to the metric in Lemma 3.39 afterwards.

One might already wonder why it isn't just possible to only pick points not having 0 in any of their coordinates to have a large set of points S that runs foul of (ii). If we picked all the points of this form, we would have $(2M-1)^N$ points in S . Note that, letting $C = N - 2M \log \ell > 0$, we have

$$\begin{aligned} (2M-1)^N &= (2M-1)^{2M \log \ell + C} \\ &= (2M)^N \left(1 - \frac{1}{2M}\right)^{2M \log \ell + C} \\ &\leq (2M)^N e^{-\log \ell + C} \\ &\leq \frac{1}{\ell} (2M)^N \end{aligned}$$

and we would not have enough points.

Proof. (Lemma 3.39) Let us firstly define the following relation: we say for $x, y \in [\mathbb{Z}/2M\mathbb{Z}]^N$, $x \sim_s y$ if $x_1 = y_1, \dots, x_s = y_s$ and $d(x_{s+1}, y_{s+1}) \leq 1, \dots, d(x_N, y_N) \leq 1$ - in other words they coincide on the first s coordinates and differ by at most 1 on the remaining coordinates. Clearly, this relation is reflexive and symmetric. Additionally, we have the chain of implications

$$x = y \iff x \sim_N y \implies x \sim_{N-1} y \implies \dots \implies x \sim_1 y. \quad (3.20)$$

Let us therefore introduce $S_n := \{y \in [\mathbb{Z}/2M\mathbb{Z}]^N : \exists x \in S \text{ s.t. } x \sim_n y\}$. The chain of implications in (3.20) induces the chain of inclusions

$$S = S_N \subseteq S_{N-1} \subseteq \dots \subseteq S_0 \subseteq [\mathbb{Z}/2M\mathbb{Z}]^N. \quad (3.21)$$

Firstly, we claim that at least one of the sets S_{n+1} contains a "hyperplane in the n th coordinate". In other words, there exists some choice of coordinates $x_0, \dots, x_{n-1}, x_{n+1}, \dots, x_{N-1}$ where the points

$$\{(x_0, \dots, x_{n-1}, i, x_{n+1}, \dots, x_{N-1}) : 1 \leq i \leq 2M\} \subseteq [\mathbb{Z}/2M\mathbb{Z}]^N$$

all lie in S_{n+1} . Denote this set as $P_n(x)$, where x is any point in the set (in other words, $P_n(x)$ is all points that coincide with x on every coordinate except perhaps n). Assume this claim is not true; this would mean that for every S_{i+1} , $0 \leq i \leq N-1$, there is no $x \in S_{i+1}$ such that $P_i(x) \subseteq S_{i+1}$. Let $\varphi(x)$ therefore denote the element in $P_i(x)$ with the next lowest choice of i -coordinate (ergo, incrementing from x_{i-1} upwards and looping modulo $2M$ if necessary until we find the desired element) not present in S_{i+1} . Such an element exists because, of course, $x \in S_{i+1}$ but not all of $P_i(x)$. Note that if we lower the i -coordinate by 1 to now obtain an element $x' \in S_{i+1}$, there exists some $z \in S$ such that $z \sim_i x'$. Additionally, $\varphi(x)$ is identical to x' in all coordinates except the $i-1$ th, where they differ by 1, so we therefore have that $\varphi(x) \in S_i$. So φ is a map from S_{i+1} to $S_i \setminus S_{i+1}$.

In particular, if we fix i then we notice that whilst φ may not be injective, since it only changes a single coordinate, any element of $S_i \setminus S_{i+1}$ has at most $2M - 1$ preimages. In particular, $|S_{i+1}| \leq (2M - 1)|S_i \setminus S_{i+1}| \implies (1 + \frac{1}{2M-1})|S_{i+1}| \leq |S_i|$. This allows us to bound the size of every set as a function of S_0 , and we now have, again letting $C = N - 2M \log \ell > 0$,

$$\begin{aligned} |S_0| &\geq \left(1 + \frac{1}{2M-1}\right)^N |S| \\ &\geq \left(1 + \frac{1}{2M-1}\right)^{2M \log \ell + C} \frac{(2M)^N}{\ell} \\ &> e^{\log \ell} e^C \frac{(2M)^N}{\ell} \\ &\geq (2M)^N \end{aligned}$$

which is a contradiction as S_0 is a subset of $[\mathbb{Z}/2M\mathbb{Z}]^N$. So the claim in (3.21) is proven.

The claim in (3.21) allows us to prove Lemma 3.39 now. We now know there is an i such that there exists $x \in S_{i+1}$ with $P_i(x) \subseteq S_i$. In particular, if we change the i th coordinate to 0 (calling the element thus obtained v) or to M (calling it w), we have two elements of S_{i+1} . But if we take v' and w' as any given elements of B such that $v' \sim_{i+1} v$ and the same with w, w' , then v' and w' coincide over the first s coordinates; v_s and w_s are equal to 0 and M respectively, and they differ by at most 2 over the remaining coordinates (due to both differing by at most 1 from v and w which are equal over these coordinates). $\square$

We will also need a strengthening of Theorem 2.17.

Theorem 3.40 *For all measure-preserving dynamical systems $(X, \mathcal{B}, \mu, T)$, and any $A \in \mathcal{B}$ such that $\mu(A) > 0$, there exists $n \leq \frac{2}{\mu(A)}$ such that*

$$\mu(A \cap T^{-n}A) \geq \frac{\mu(A)^2}{2}.$$

Proof. For every $n \in \mathbb{N}$, we define $f_n: X \rightarrow \mathbb{R}$ as $f_n := \mathbb{1}_{T^{-n}A}$. Let $x = \mu(A)$ and take $N \in \mathbb{N}$ with $N \geq \frac{2}{x}$. Let

$$y = \max_{i \neq j \in [N]} \mu(T^{-i} \cap T^{-j}A).$$

We want to show that $y \geq \frac{x^2}{2}$. Using Cauchy-Schwarz, we have that

$$\begin{aligned}
 (Nx)^2 &= \left(\int_X \sum_{n=1}^N f_n d\mu \right)^2 \stackrel{\text{CS}}{\leq} \int_X \left(\sum_{n=1}^N f_n \right)^2 d\mu \\
 &= \sum_{n=1}^N \int_X f_n^2 d\mu + 2 \sum_{n=1}^{N-1} \sum_{m=n+1}^N \int_X f_n f_m d\mu \\
 &= Nx + \sum_{n=1}^{N-1} \sum_{m=n+1}^N \mu(T^{-n}A \cap T^{-m}A) \\
 &\leq Nx + (N^2 + N)y.
 \end{aligned}$$

Since $0 \leq x \leq 1$, we have $x^2 \leq x$. So it follows that $(N^2 + N)y \geq N^2x^2 - Nx \geq (N^2 - N)x^2 \implies y \geq \frac{x^2}{2}$. $\square$

Using these two results, we can prove another combinatorial lemma. The usefulness of all these results may seem obfuscated for now, but as our eventual proof will be very constructive, it will make sense shortly.

Lemma 3.41 *Let $\ell, M \in \mathbb{N}$ with M taken sufficiently large depending on the value of ℓ . Take $N \in \mathbb{N}$ such that $N \geq 2M \log 2\ell^2$. Then for any set $S \subseteq [(2M)^N]$ with $|S| > \frac{1}{\ell}(2M)^N$, there is a positive integer $y \in (S - S)$ such that when y is written in base $2M$:*

$$y = \sum_{n=0}^{N-1} y_n (2M)^n \text{ s.t. } 0 \leq y_n \leq 2M - 1$$

then one of the digits y_i is between M and $M + 8\ell$ and the other digits are at most 8ℓ .

The setup is very similar to that in Lemma 3.39 with the caveat that we are now in the positive integers rather than $[\mathbb{Z}/2M\mathbb{Z}]^N$ so the distance does not “loop around”. Lemma 3.39 tells us that there are two points whose coordinates are equal up to a coordinate, equal to 0 and M in that coordinate and differ by at most 2 after. But our convention on distance meant that $2M - 1$ and 0 differed by 1, which does not translate well to the integers. We therefore need to adapt this aspect of the lemma to our proof here.

Proof. (Lemma 3.41) Let $\ell, 2M, N, S$ be taken as described in the statement of Lemma 3.41, and let ψ denote the canonical (bijective) mapping from $[(2M)^N]$ to $[\mathbb{Z}/2M\mathbb{Z}]^N$ sending a nonnegative integer to the N -uple of the digits of its base- $2M$ expression (and as an abuse of notation, for any $S \subseteq [(2M)^N]$, let $\varphi(S) := \{\varphi(s) : s \in S\} \subseteq [\mathbb{Z}/2M\mathbb{Z}]^N$). For any $x \in [(2M)^N]$ or $x \in [\mathbb{Z}/2M\mathbb{Z}]^N$ (it will be clear which case we are in at any given time), x_i denotes the i th coordinate of x (when written in base $2M$ in the former case, or just the coordinate outright in the latter case). Evidently, $x_i = \varphi(x)_i$. We

also consider the (uniform, discrete) probability measure μ where

$$\mu(X) = \frac{1}{(2M)^N} |X|$$

for any subset $X \subseteq [\mathbb{Z}/2M\mathbb{Z}]^N$ and note that any bijective mapping is μ -measure-preserving. In particular, the mapping T given by

$$T: [\mathbb{Z}/2M\mathbb{Z}]^N \rightarrow [\mathbb{Z}/2M\mathbb{Z}]^N, T(x_0, \dots, x_{N-1}) = T(x_0 + 4, \dots, x_{N-1} + 4)$$

is a measure-preserving transformation, allowing us to apply our results from ergodic theory.

Firstly, $\mu(\varphi(S)) \geq \frac{1}{\ell}$ by assumption, so applying Theorem 3.40 we see that there is some $n \in \mathbb{N}$ less than $\frac{2}{\mu(\varphi(S))}$ (which is in turn bounded by $2j$) such that $\mu(T^{-n}\varphi(S) \cap \varphi(S)) \geq \frac{\mu(\varphi(S))^2}{2}$. In particular, $|T^{-n}\varphi(S) \cap \varphi(S)| = \mu(T^{-n}S \cap S)(2M)^N \geq \frac{1}{2\ell^2}(2M)^N$ allowing us to invoke Lemma 3.39 by taking $\ell' = 2\ell^2$ in the statement of the lemma. This yields $x, y \in T^{-n}\varphi(S) \cap \varphi(S)$ such that there exists $0 \leq n \leq N-1$ with

- (i) $x_i = y_i$ for $0 \leq i \leq s-1$;
- (ii) $x_s = 0, y_s = M$;
- (iii) $d(x_i, y_i) \leq 2$ for $s+1 \leq i \leq N-1$.

Note also that $T^{-n}(y)$ and x are in $\varphi(S)$. The s th coordinate of $T^n(y)$ is equal to $M + 4n$, and $n \leq 2j$ by assumption. Since we are taking $2M$ as a sufficiently large value as a function of j , we can take it as a function of n as well to ensure $M + 4n < 2M$, which is the s th coordinate of x . So we can actually ensure that $T^{-n}(y), x$ are two distinct points by increasing Q .

Finally, we return to the integers by taking $\varphi^{-1}(x), \varphi^{-1}(T^n(y))$ instead. We wish to show that these satisfy the desired properties in the statement of the lemma. Let $z = |\varphi^{-1}(x) - \varphi^{-1}(T^n(y))|$. As mentioned, when subtracting there might be an issue due to no longer being able to loop around (ergo, borrowing 1 from the next higher digit) but this affects the result by at most 1. As a result, we know that

$$z_i \bmod 2M \in \{y_i + 4n - x_i, y_i + 4n - x_i - 1, x_i - y_i - 4n, x_i - y_i - 4n - 1\}$$

and all four of these are at most $8j$, save for the specific case of z_s , which is instead between M and $M + 8j$ as desired. $\square$

Lastly, we need to establish the existence of measures with certain properties that we outline below in Lemma 27. We will only state how these measures are constructed; for full details of why the properties in Lemma 27 hold, the reader can consult Mountakis' paper[10].

Lemma 3.42 *Let $\ell, M \in \mathbb{N}$ with $2M$ taken sufficiently large depending on ℓ . Then there is a positive constant C independent of the choice of $\ell, 2M$ (that we will determine later) such that for every $k \in \mathbb{N}$, there is a non-negative finite measure σ on $\mathbb{T}$ such that*

- (i) σ is supported on the $(2M)^k$ th roots of unity; i.e. the total measure of the points $0, \frac{1}{(2M)^k}, \dots, \frac{(2M)^k-1}{(2M)^k}$ is 1. Note that this implies that

$$\widehat{\sigma}(k) = \widehat{\sigma}(k + (2M)^k)$$

via a simple calculation.

- (ii) $\widehat{\sigma}(0) \leq 1 + C \frac{\ell^3}{(2M)^2}$.
- (iii) $\widehat{\sigma}(m) = 1$ for $(2M)^k \leq m \leq \ell(2M)^k$.
- (iv) $\widehat{\sigma}(m) = -1$ for $M(2M)^k \leq m \leq (M + \ell)(2M)^k$.

Proof. (Proof sketch of constructive part of Lemma 27) We can assume $M > 2\ell$ as we are free to take it as large as desired as a function of ℓ . Define $N := (2M)^{k+1}$. We now define trigonometric polynomials which have desirable properties and will allow us to construct σ as described in Lemma . Let

$$P(x) := \sum_{m=-\ell(2M)^k}^{\ell(2M)^k} \left(1 - \cos 2\pi \frac{\ell(2M)^k - |m|}{N} \right) e(mx)$$

and

$$R(x) := 2 \cos(\ell(2M)^k 2\pi x) - \cos \left(\left(\frac{N}{2} - \ell(2M)^k \right) 2\pi x \right) - \cos \left(\left(\frac{N}{2} + \ell(2M)^k \right) 2\pi x \right)$$

with both polynomials defined on $\mathbb{T}$; note that both take real values uniquely although we will not prove this here. Recall also that $F_{(2M)^k}$ denotes a Fejér kernel and define

$$S(x) = \left(16\ell * F_{(2M)^k} \right) (x) + R(x)P(x).$$

We want to show that $S \geq 0$; it will be useful to apply Lemma 3.37 here. Indeed, if we consider the sequence $(x_i)_{i=0}^\infty$ where

$$x_i = 1 - \cos \left(2\pi \frac{\ell(2M)^k - i}{N} \right) \text{ for } 1 \leq i \leq \ell(2M)^k$$

and $x_i = 0$ otherwise then by our assumption $4\ell < 2M$ we have that all the values of $\frac{\ell(2M)^k - i}{N}$ lie between 0 and $\frac{\pi}{2}$, over which the function $f(x) = 1 - \cos(2\pi(\frac{\ell(2M)^k - i}{N}))$ is decreasing and convex, implying the same for the sequence up to $x_{\ell(2M)^k}$ (and it remains decreasing and convex, as it is

constantly equal to 0 afterwards). This allows us to apply Lemma 3.37 and as a result $p(x)$ is nonnegative. We can then apply Lemma 3.36 to further show that

$$P(x) \leq \left[(4\ell P) * F_{(2M)^k} \right] (x).$$

We also note that $R(x)$, as the sum of 4 cosine functions, is bounded in absolute value by 4. This gives us that

$$-R(x)P(x) \leq |R(x)|P(x) \leq 4P(x) \left[(4\ell P) * F_{(2M)^k} \right] (x)$$

proving the desired positivity of S .

Now, we define a measure σ on $\mathbb{T}$ as

$$\sigma = \frac{1}{2} \left(\delta_{\frac{1}{N}} + \delta_{\frac{N-1}{N}} \right) + \frac{1}{N} \sum_{n=0}^{N-1} s \left(\frac{n}{N} \right) \delta_{\frac{N-n}{N}}$$

with the standard definition: δ_x being the Dirac measure at x (the measure equal to 1 for sets containing x and 0 otherwise). Note that σ is not a probability measure but it is indeed finite as the sum of finitely many Dirac measures. It is also supported on the N th roots of unity. We claim that this measure has exactly the properties described in the statement of Lemma 27. $\square$

Without showing the full proof we cannot explain which constant would work, but if one follows the calculations in Mountakis' paper, for example, then the value $C = 320$ is obtained.

We are finally ready to conclude the proof of Theorem 3.38. Recall briefly what we want to show: for every $j \in \mathbb{N}, \varepsilon \in]0, \frac{1}{2}[$, the existence of a finite set U_j that is a set of $\frac{1}{j}$ -recurrence but not an ε -vdC set. Let us begin with a brief refresher of what we are looking to achieve.

In order to prove the theorem, we are going to create a measure μ using Lemma 27 such that $\mu(\{0\}) > \varepsilon$ and then consider the set

$$U_j = \{n \in \mathbb{N} : \widehat{\mu}(n) = 0\}$$

which is, of course, not an ε -vdC set - but which will turn out to be a set of $\frac{1}{j}$ -recurrence using Lemma 3.41.

Let us think about how the latter part can be achieved. From our previous results (Lemma 3.39, Lemma 3.41), we are motivated to work with $[(2M)^k - 1]$ for some k and show that for any subset $S \subseteq [(2M)^k - 1]$ with density at least $\frac{1}{j}$, the difference set $S - S$ intersects with U_j . Now, Lemma 3.41 already tells us that with a sufficiently large k and $2M$, we can find an element y of the difference set with the property that when it is written in base $2M$, the all the coefficients but one are bounded by $8j = \ell$, the last coefficient being

between M and $M + \ell$. How do we construct U_j - or rather, μ - in such a manner that $\widehat{\mu}(y) = 0$?

Lemma 27 gives us a measure as a function of some number k - call it σ_k - with Fourier coefficients 1 or -1 on a large set, supported on $(2M)^{k+1}$ th roots of unity and with the additional property that $\widehat{\sigma}_k(0)$ is bounded. The key idea will now be to create a measure σ that is the convolution of many such measures σ_k for different values of k . Recall that the Fourier transform of the convolution is the product of Fourier transforms, which is very useful. The aim will be to do this in such a manner that

- The properties bestowed on y via the invocation of Lemma 3.41 (its digits being either in a specific range or very small) along with the Fourier transform of σ_k being 1 or -1 on a corresponding ranges to the properties of y will allow us to conclude that all the terms in the resultant product are 1 or -1 , leading to $\widehat{\sigma}(y) = -1$;
- We can find a good bound on the total measure $\sigma(\mathbb{T})$.

We will then "normalize" the measure σ , but only after first adding a point mass of 1 at $\{0\}$. This will ensure that knowing $\widehat{\sigma}(y) = -1$ leads to $\widehat{\mu}(y) = 0$.

We proceed now to the details of the proof of Theorem 3.38.

Proof. (Theorem 3.38) We choose $M \in \mathbb{N}$ with $2M$ sufficiently large, and define $\ell = 8j$ to then apply Lemmas 3.41 and 27. For reasons of clarity, we will assume from here forward that $2M$ has any desired property we run into and then summarise at the end what properties we need and why they are all possible.

Let $N \in \mathbb{N}$ be a large number in function of M (and by extension, j) that we will pick later, and for each $k \in \{0, 1, \dots, N-1\}$, apply Lemma 27 with $\ell = 8j$ to obtain a measure σ_k . Let C be the constant from Lemma 27. Take

$$\sigma = \sigma_0 * \dots * \sigma_{N-1}$$

For all of these, we know that $\widehat{\sigma}_k(0) \leq 1 + C \frac{\ell^3}{(2M)^2}$, so (not forgetting $\ell = 8j$) it follows that, by our earlier remark on how the Fourier transform of the convolution is the product of Fourier transforms,

$$0 \leq \sigma(\mathbb{T}) = \widehat{\sigma}(0) = \widehat{\sigma}_0(0) \dots \widehat{\sigma}_{N-1}(0) \leq \left(1 + C \frac{\ell^3}{(2M)^2}\right)^N.$$

Let δ_0 denote the point mass of 1 at 0. Take μ to be the normalisation of the measure $\sigma + \delta_0$; namely, $\mu := \frac{\sigma + \delta_0}{\sigma(\mathbb{T}) + 1}$. Clearly, μ is a probability measure. Now we can calculate the measure of $\{0\}$, which is

$$\mu(\{0\}) = \frac{\sigma(\{0\}) + 1}{1 + \sigma(\mathbb{T})} \geq \frac{1}{1 + \sigma(\mathbb{T})} \geq \frac{1}{1 + \left(1 + C \frac{\ell^3}{(2M)^2}\right)^N}. \quad (3.22)$$

We want the last expression in (3.22) to be bigger than ε . In particular, (combining the constants together into a single term C_1 to make things simpler to quantify) the expression

$$\left(1 + \frac{C_1}{M^2}\right)^N \quad (3.23)$$

is decreasing as a function of $2M$. Beware, though - N must stay large as a function of $2M$ - in particular, when we wish to apply Lemma 3.41, so we will need that $N > 2M \log 2j^2$. However, if we just take $N = \lfloor 2M \log 2j^2 \rfloor + 1$ and re-evaluate (3.23), which is now (again combining constants into C_1 and C_2 , this time with a small leap mentioned below)

$$\left(1 + \frac{C_1}{M^2}\right)^{MC_2}$$

is still decreasing as a function of $2M$ - indeed, we know that

$$\left(1 + \frac{C_1}{M^2}\right)^{MC_2} = \left[\left(1 + \frac{C_1}{M^2}\right)^{M^2 C_2}\right]^{\frac{1}{M}} \leq [e^{\frac{C_1}{C_2}}]^{\frac{1}{M}}$$

which can notably be brought as close to 1 as we like by choosing an appropriate M . We ignored a small asymptotic effect - a $+1$ in the exponent - but we can remove a constant C_3 from the exponent and only divide by $\left(1 + \frac{C_1}{M^2}\right)^{C_3}$, which goes towards 1 as well. We therefore choose a suitable M to then get that

$$\frac{1}{1 + \left(1 + C \frac{\ell^3}{(2M)^2}\right)^N}$$

is arbitrarily close enough to $\frac{1}{2}$ (but smaller) but larger than ε . It follows that the set

$$U_j = \{n \in \mathbb{N} : \widehat{\mu}(n) = 0\}$$

is not an ε -vdC set. This will not be the only constraint we need on M - but both are lower bounds, so we take M sufficiently large to satisfy both. We will note the other constraint on M shortly.

Let us proceed to the second part - showing that U_j is a set of $\frac{1}{j}$ recurrence. As explained previously, we will apply Lemma 3.39 with any subset $S \subseteq [(2M)^N - 1]$ having density at least $\frac{1}{j}$ to obtain a positive integer $y \in (S - S)$ and $1 \leq s \leq N - 1$ such that when y is written in base $2M$ we have that

- (i) $M \leq y_s \leq M + 8j$;
- (ii) for $i \neq s$, $y_s \leq 8j$.

Recall that $\widehat{\sigma}_k$ is periodic with period $(2M)^{k+1}$ which implies

$$\widehat{\sigma}(y) = \prod_{k=0}^{N-1} \widehat{\sigma}_k(y) = \prod_{k=0}^{N-1} \widehat{\sigma}_k \left(\sum_{i=0}^{N-1} y_i 2M^i \right)$$

In particular, if $i > k$ then $\widehat{\sigma}_k(t + y_i(2M)^i) = \widehat{\sigma}_k(t)$ since $(2M)^{k+1} \mid (2M)^i$ and so we can ignore all the terms in the summation for $i > k$.

Here, we ensure $2M \geq 8j$ so that $\frac{8j-1}{2M-1} \leq 1$. We already know that

$$M \leq \frac{\sum_{i=0}^s y_i (2M)^i}{(2M)^s}$$

but we can now provide an upper bound; in particular:

$$\frac{\sum_{i=0}^s y_i (2M)^i}{(2M)^s} \leq M + 8j - 1 + \frac{8j-1}{(2M)^s} \sum_{i=0}^{s-1} y_i (2M)^i \leq M + 8j$$

In particular, this now means that $\sigma_s(y) = -1$ because of the properties bestowed upon these measures by Lemma 27. Furthermore, for any $k \neq s$ we have

$$1 \leq \frac{\sum_{i=0}^k y_i (2M)^i}{(2M)^k} = y_k + \sum_{i=0}^{k-1} y_i (2M)^{i-k} \leq 8j$$

and $\sigma_k(y) = 1$. Plugging these into the product we have $\sigma(y) = 1$ and (by the additivity of the Fourier transform)

$$\widehat{\mu}(y) = \frac{\widehat{\sigma}(y) + \widehat{\delta}_0(y)}{\sigma(\mathbb{T}) + 1} = 0$$

meaning that $y \in U_j$ as desired. This means U_j is indeed a set of $\frac{1}{j}$ -recurrence as desired.

One last finishing touch: of course, U_j is not finite. But if we simply now take $U_j \cap [(2M)^N]$ then nothing changes in the proof. So we are done. $\square$

With this result under our belt, we are very close to proving the main result of Mountakis' paper. Mountakis' proof actually improves on the first paper by Bourgain - it provides a set of *strong recurrence*.

Definition 3.43 (Strong recurrence) A set $R \subseteq \mathbb{N}$ is a set of strong recurrence if, given any measure-preserving system $(X, \mathcal{B}, \mu, T)$ and any set $A \in \mathcal{B}$ with $\mu(A) > 0$, we have

$$\limsup_{n \rightarrow \infty, n \in R} \mu(A \cap T^{-n}A) > 0.$$

It is clear that all sets of strong recurrence are sets of recurrence.

It was already remarked that ε -vdC sets are van der Corput sets so it suffices to show the following, in order to establish the existence of a set of strong recurrence that is not a van der Corput set.

Theorem 3.44 (Mountakis) *For every $\varepsilon \in]0, \frac{1}{3}[$ there is a set of strong recurrence that is not an ε -vdC set.*

Proof. In order to use Theorem 3.38 to prove Theorem 3.44, we will need to find an argument to somehow "combine" a sequence of finite $\frac{1}{j}$ -recurrence sets into a set of strong recurrence.

Take any $\varepsilon \in]0, \frac{1}{3}[$ and let $\xi \in]0, \frac{1}{2}[$ be any value satisfying $\frac{\xi}{1+\xi} \geq \varepsilon$. We can therefore, via Theorem 3.38, find finite sets $S_1, S_2, \dots \subseteq \mathbb{N}$ such that S_i is a set of $\frac{1}{i}$ -recurrence but not an ξ -vdC set - in particular, by definition, there is a probability measure μ_i such that

$$S_i \subseteq \{n \in \mathbb{N} : \hat{\mu}_i(n) = 0\}; \mu(\{0\}) \geq \xi.$$

Let $n_i \in \mathbb{N}$ denote any value for which S_i satisfies the conditions of the definition of $\frac{1}{i}$ -recurrence; namely for any subset E of $[n_i]$ such that

$$|E| \geq \frac{n_i}{i}$$

we have $(E - E) \cap S_i \neq \emptyset$. As mentioned at the end of the previous proof, if S_i is a set of $\frac{1}{i}$ -recurrence then so is $S_i \cap [n_i]$ and so we may assume that $\max(S_i) \leq n_i$, by taking the aforementioned restriction if necessary. Now, consider the sequence

$$T_{\frac{k(k+1)}{2}+j} := S_j$$

where every positive integer n is written as $\frac{k(k+1)}{2} + j$ in the unique manner satisfying $1 \leq j \leq k$; in other words, the sequence given by

$$S_1, S_1, S_2, S_1, S_2, S_3, \dots$$

that contains every S_i infinitely many times. The idea is that returning to S_i in this manner will allow us to ensure strong recurrence later on. We let $\chi: \mathbb{N} \rightarrow \mathbb{N}$ denote the mapping sending $n = \frac{k(k+1)}{2} + j$: $1 \leq j \leq k$ to j , and obviously have that $T_k = S_{\chi(k)}$. Now fix k , and define the measure $\sigma_k = \mu_{\chi(k)} - \xi\delta_0$ where we recall δ_0 is the measure assigning point mass 1 to $\{0\}$. We then have that the sequence of measures $(\sigma_i)_{i=0}^\infty$ consists only of nonnegative measures (but no longer probability measures).

We will now endeavour to construct a positive trigonometric polynomial with certain properties. Namely, taking $m_j := \frac{1}{\xi}n_j(n_j + 1)$, we want to find a polynomial P_j such that

- (i) $\widehat{P}_j(m) = 0$ for $m \in \mathbb{Z}$ with $|m| > m_j$;
- (ii) $\widehat{P}_j(m) = \widehat{\sigma}_j(m)$ for $|m| \leq n_j$ with the exception of 0;
- (iii) $\widehat{P}_j(0) = 1$.

Now whilst we could simply define our function as being the trigonometric polynomial with specifically these coefficients (although not all the coefficients are given), this would not prove positivity. We will therefore use the positivity of a similar-looking function already seen before to help: the Fejér kernel.

We already know the Fourier coefficients of the Fejér kernel, and they have the nice property of being eventually 0. If we consider the function $F_{m_j} * \sigma_j$ then since we know $\widehat{F_{m_j} * \sigma_j} = \widehat{F_{m_j}} \widehat{\sigma_j}$,

$$\widehat{F_{m_j} * \sigma_j}(m) = \begin{cases} 0 & \text{if } |m| \geq m_j \\ \widehat{\sigma_j}(n) \left(1 - \frac{|m|}{m_j}\right) & \text{if } |m| \leq m_j \end{cases}.$$

This is a non-negative trigonometric polynomial by the non-negativity of both F_{m_j} and σ_j . In order to get to a polynomial satisfying (i), (ii) and (iii), we simply need to add a polynomial that "completes" the Fourier coefficients between $-n_j$ and n_j , whilst not affecting those lesser than $-m_j$ or greater than m_j . For example, if we take a function $a_j: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\widehat{a_j}(m) = \begin{cases} 0 & \text{if } |m| \geq n_j \\ \widehat{\sigma_j}(m) \frac{|m|}{m_j} & \text{if } |m| \leq n_j \end{cases}$$

then it is clear that $a_j + F_{m_j} * \sigma_j$ satisfies all of the desired properties except perhaps positivity, which remains to be shown. As a matter of fact, the positivity of a_j is not guaranteed - but we can show it is bounded from below, which is enough (since shifting the function up by a constant does not change its Fourier coefficients). Indeed,

$$\min_{x \in \mathbb{R}} a_j(x) = \min_{x \in [0,1]} a_j(x).$$

This is since, like all trigonometric polynomials, a_j is periodic with period 1 (see Section 2.3). and we have the following inequality for all $x \in [0, 1]$:

$$|a_j(x)| \leq \sum_{n \in \mathbb{N}} |\widehat{a_j}(m)| = \sum_{n=-n_j}^{n_j} |\widehat{a_j}(m)| = \sum_{n=-n_j}^{n_j} |\widehat{\sigma_j}(m)| \frac{|m|}{m_j} < \sum_{n=-n_j}^{n_j} \frac{|m|}{m_j}$$

and by our constraint on m_j this sum is at most ξ . So the sum $a_j + F_{m_j} * \sigma_j + \xi$ is a trigonometric polynomial satisfying everything we need. Take such a trigonometric polynomial for every $j \in \mathbb{N}$ and call it b_j . We now define two

3.3. Sets of recurrence that are not van der Corput

more sequences of polynomials $(c_i)_{i=0}^\infty$ and $(d_i)_{i=0}^\infty$ as well as a sequence of positive integers $(k_i)_{i=0}^\infty$. Let $c_1 := b_1$, $k_1 := m_1$, and then define subsequently

$$\begin{aligned} k_j &= 2(m_{j-1} + 1)k_{j+1} + 1; \\ d_j(t) &= b_j(2k_j t); \\ c_j &= c_{j-1}(t)d_j(t). \end{aligned}$$

By the positivity of b_j , we have that d_j and c_j are positive polynomials as well. Solving for the Fourier coefficients and rearranging, we also have

$$\widehat{d_j}(2k_j m) = \widehat{b_j}(m), \widehat{d_j}(i) = 0 \text{ for all other input values } i.$$

Lemma 3.45 *The Fourier coefficients $\widehat{c_j}(i)$, $i \in \mathbb{Z}$ have the following properties:*

- (i) $\widehat{c_j}(m) = 0$ for $|m| \geq k_{j+1}$
- (ii) $\widehat{c_j}(m) = \widehat{c_{j+1}}(m)$ for $|m| < k_{j+1}$;
- (iii) $\widehat{c_j}(0) = 1$;
- (iv) $\widehat{c_j}(2k_j m) = -\delta$ for $m \in S_j$.

Rigorous justification of Lemma 3.45 can be found in Mountakis' paper[10] on page 12.

Now, by Lemma 3.45, we know c_j is nonnegative, Borel measurable, and

$$\int_{\mathbb{T}} c_j(t) dt = \widehat{c_j}(0) \tag{3.24}$$

by definition, and we know the RHS in (3.24) is equal to 1 from (iii). As a result, we can induce a probability measure on $\mathbb{T}$ by taking $\mu_j(t) := c_j(t)$ for $t \in \mathbb{T}$ (since c_j is periodic, there are no issues with discontinuities). By Bolzano-Weierstrass and the weak* compactness of probability measures on $\mathbb{T}$, there is a convergent subsequence whose limit we call μ . We again wish to repeat the trick of ensuring that there is a point mass of value ξ at 0 and then using this to generate a set that is not ξ -vdC. We therefore add the aforementioned point mass and normalize to obtain a probability measure ρ :

$$\rho := \frac{\mu + \xi \delta_0}{\mu(\mathbb{T}) + \xi \delta_0(\mathbb{T})} = \frac{1}{1 + \xi} \mu + \frac{\xi}{1 + \xi} \delta_0.$$

Let $R := \{R \subseteq \mathbb{N} : \widehat{\rho}(r) = 0\}$. Of course, by construction, R is not ξ -vdC. Additionally, by construction, R contains $2k_j S_j$ for any choice of $j \in \mathbb{N}$ (with the abuse of notation that for $n \in \mathbb{N}$, $S \subseteq \mathbb{N}$, $nS := \{ns : s \in S\}$). This is easy to see: first, we may assume that the subsequence of $(\mu_j)_{j=1}^\infty$ converging to μ is the entire sequence (if it were not, we could extract the subsequence and

work solely on that, remapping to $\mathbb{N}$). For a fixed j and for a fixed $i \in S_j$, and $\ell > j$ we have that

$$\widehat{c_{\ell-1}}(2k_j i) = \cdots = \widehat{c_j}(2k_j i)$$

by our properties of c_j (using that $2k_j i \leq 2k_j \max(S_j) \leq 2k_j n_j \leq N_\ell$). In particular,

$$\widehat{\mu}(2k_j i) = -\delta.$$

Calculating now, we can see for any $2k_j i$, with $i \in R_j$ again, that $\widehat{\rho}(2k_j i) = 0$.

To finish, we will now show that the union

$$\bigcup_{j \in \mathbb{N}} 2k_j R_j$$

is a set of strong recurrence, and we will be done by the upwards closure of the family of sets of strong recurrence.

Take any measure-preserving dynamical system $(X, \mathcal{B}, \mu', T)$ and $A \in \mathcal{B}$ with $\mu'(A) > 0$. Now, if we take any integer i greater than $\frac{2}{\mu'(A)}$, we remark the following.

Let $j \in \chi^{-1}i$, and let S be the measure-preserving transformation given by T^{2k_j} . For every $x \in X$, we denote by $V(x)$ the set of "visits" to A via S that are at most n_j :

$$V(x) := \{n \leq n_j : S^n x \in A\}.$$

Then we can define a function v as follows.

$$v(x) = \frac{1}{n_j} |V(x)|.$$

This function represents how often we visit A . Now, as T is measure-preserving, so is S . So in particular, since the measure of A is greater than $\frac{2}{i}$ by assumption, it follows that the integral $\int_X v(x) d\mu'(x) \geq \frac{2}{i}$ as well since the LHS happens to be equal to exactly $\mu'(A)$. This means that if we let W_j be the set $\{x \in X : |V(x)| \geq \frac{n_j}{i}\}$ then $\mu'(W_j) > \frac{1}{i}$. We have 2^{n_j} different ways for taking a subset of $[n_j]$ (one of which is $V(x)$) so pigeonholing over all the possible subsets, and noting that $V(x)$ must be exactly one of them, there is some subset $E \subseteq [n_j]$ where $|E| \geq \frac{1}{i} n_j$ and

$$\mu'(\{x \in X : V(x) = E\}) \geq \frac{1}{i 2^{n_j}} \quad (3.25)$$

as the union of all such sets is W_j . We then denote the set $\{x \in X : V(x) = E\}$ from the LHS of (3.25) as C . As $|E| \geq \frac{1}{i} n_j$, then by the definition of $\frac{1}{i}$ -recurrence, $E - E$ contains an element of S_i . Let n be any such element, and by our construction we have $r \in E$ such that $n, n + r$ are both in E . As a

3.3. Sets of recurrence that are not van der Corput

consequence, C must be a subset of $T^{-n}(A \cap T^{-r}A)$, which of course has the same measure as $(A \cap T^{-r}A)$. So

$$\frac{1}{i2^{n_i}} \leq \mu'(A \cap T^{-r}A) = \mu'(A \cap S^{-2k_j r}(A))$$

proving that, since $r \in S_i$, and $\chi(j) = i$, we have infinitely many shifts for which the measure of the intersection is above a certain threshold - the definition of strong recurrence. This concludes the proof. $\square$

Chapter 4

Roth's theorem on arithmetic progressions in different contexts

In this chapter, we will provide two proofs of Roth's theorem on arithmetic progressions - a combinatorial proof, due to Szemerédi, and an ergodic proof, due to Bergelson. The latter establishes a general form of Roth's theorem in arbitrary countable abelian groups.

4.1 A combinatorial proof of Roth's theorem

We start with formulating the classical Roth theorem[3] on 3-term arithmetic progressions.

Theorem 4.1 (Roth's theorem) *Any $S \subseteq \mathbb{N}$ such that $\bar{d}(S) > 0$ contains a 3-term arithmetic progression.*

We will refer to a subset of the integers not containing a 3-term arithmetic progression as a *progression-free set*.

4.1.1 The equivalence of finitary and infinitary Roth

Like many results in additive combinatorics, Roth's theorem also has an equivalent finitary version.

Theorem 4.2 (Finitary Roth's theorem) *Given $N \in \mathbb{N}$, let $r_3(N)$ be the size of the largest subset of $[N]$ that is a progression-free set. Then $\lim_{N \rightarrow \infty} \frac{r_3(N)}{N} = 0$.*

Lemma 4.3 *Theorem 4.1 and Theorem 4.2 are equivalent.*

Proof. $4.2 \implies 4.1$: If S is an subset of $\mathbb{N}$ with positive upper density, then there must be some $N \in \mathbb{N}$ for which

$$\frac{|S \cap [N]|}{N} > \frac{r_3(N)}{N}$$

4.1. A combinatorial proof of Roth's theorem

as the limsup of the LHS (as $N \rightarrow \infty$) is greater than 0, whilst the limit of the RHS is 0. By the definition of $r_3(N)$, $S \cap [N]$ is not a progression-free set, and so neither is S .

4.1 $\implies$ 4.2: Let us assume that the finitary version does not hold and construct a subset S of the natural numbers with positive upper density containing a 3-term arithmetic progression, much like in the argument in Section 2 with syndetic sets.

By assumption, there exists $\varepsilon > 0$ such that for all $N \in \mathbb{N}$, there exists an integer $M > N$ and a subset $S_M \subseteq [M]$ such that S_M is a progression-free set and

$$\frac{|S_M \cap [M]|}{M} > \varepsilon. \quad (4.1)$$

In particular, we have an infinite number of such sets, each satisfying (4.1) over progressively longer intervals $M_1, M_2, \dots$; we call these sets $S_{M_1}, S_{M_2}, \dots$.

We can try collating the sets together in order to create S . Of course, writing the first M_1 terms of S_{M_1} , followed by the first M_2 terms of S_{M_2} and so forth will not work, as we might have arithmetic progressions spanning two sets. However, if we now write the first M_1 terms of S_{M_1} , followed by M_2 zeroes, followed by the first M_2 terms of S_{M_2} , followed by M_3 zeroes and so forth, then this solves this issue - if the first two terms of an arithmetic progression are in the same interval corresponding to S_{M_i} , then the third one will either forcibly be in that same interval or in the large section of zeroes inserted after, which are both contradictions. We should not ignore that there might be arithmetic sequences whose first two terms are in two different subintervals corresponding to M_i, M_j but then by taking $M_j \geq 3M_{j-1}$, we can ensure that the third term will land in some section of zeroes.

The resulting sequence M has upper density at least $\frac{\varepsilon}{2}$ and is progression-free, contradicting Theorem 4.1.

As shown in the diagram below, an arithmetic progression contained entirely in a section M_i is impossible. If we instead take the first element x in M_i and the second y in M_j , $j > i$, then since $M_j < y - x < 2 \cdot \sum_{n=1}^j 2M_n < M_{j+1}$, the difference between the two will always be enough to enter the next section of zeroes but not enough to cross it entirely. The inequality $\sum_{n=1}^j 2M_n < M_{j+1}$ is a direct consequence of taking $M_j \geq 3M_{j-1}$ at each step.

4.1. A combinatorial proof of Roth's theorem

This graph has the notable property that any triangle x_i, y_j, z_k induces an arithmetic progression $j - i, \frac{k-i}{2}, k - j$ in A . It follows that since A is progression-free, these three values must be equal. This in turn means that i, j, k (in this order) is an arithmetic progression, albeit $\pmod{2N+1}$. But this means that choosing any two of these values uniquely determines the third, so every edge is in exactly one triangle. By Theorem 4.4, there are $o(N^2)$ edges, but the number of edges is also equal to $(6N+3)|A|$, implying $|A| = o(N)$ as desired.

Let us proceed now to the proof of the theorem.

Proof. (Theorem 4.4) We begin by introducing Szemerédi's regularity lemma[4]. We start with some pertinent definitions.

Definition 4.5 (Edge density) *The edge density between two disjoint subsets $X, Y \subseteq V$ of the vertex set of a graph G is*

$$d(X, Y) := \frac{|E(X, Y)|}{|X||Y|}$$

where $E(X, Y)$ is the set of edges having one endpoint in X and the other in Y .

This definition is the most natural way to define edge density, as it is just the proportion of edges present over the maximum number there could be.

Definition 4.6 (ϵ -regularity) *Let $\epsilon > 0$. A pair (X, Y) of two disjoint subsets $X, Y \subseteq V$ of the vertex set of a graph G are said to be ϵ -regular if for all $A \subseteq X$ such that $|A| \geq \epsilon|X|$ and $B \subseteq Y$ such that $|B| \geq \epsilon|Y|$ we have*

$$|d(X, Y) - d(A, B)| < \epsilon$$

where $|E(X, Y)|$ is the set of edges having one endpoint in X and the other in Y .

In other words, ϵ -regularity just means that the edge density between sufficiently large subsets of X, Y is close to the overall edge density.

Definition 4.7 (ϵ -regularity, ctd.) *A partition of the vertex set V into k sets $\{V_1, \dots, V_k\}$ is called ϵ -regular if*

$$\sum_{(v_i, v_j) \text{ not } \epsilon\text{-regular}} |V_i||V_j| \leq \epsilon|V(G)|^2.$$

Finally, here is the statement of the regularity lemma.

Lemma 4.8 (Szemerédi's regularity lemma) *For every $\epsilon > 0, m \in \mathbb{N}$, there exists $M \in \mathbb{N}$ such that every graph with at least M vertices has a ϵ -regular partition into at least m and at most M parts.*

We will not prove this result; a proof can be found in a survey of Tao[29].

An intuitive result following from Szemerédi's regularity lemma is the Triangle counting lemma, which tells us that given a triple of vertex sets (X, Y, Z) such that all three pairs $(X, Y), (X, Z), (Y, Z)$ are ε -regular, and the edge density is sufficiently large, then there are many triangles having one vertex in each of X, Y, Z .

Lemma 4.9 (Triangle counting lemma) *Let $G = (V, E)$ be a graph and consider three disjoint subsets $X, Y, Z \subseteq V$ of the vertex set such that all three pairs $(X, Y), (X, Z), (Y, Z)$ are ε -regular. If*

$$\min(d(X, Y), d(X, Z), d(Y, Z)) \geq 2\varepsilon$$

then the number of triangles having one vertex in each of X, Y, Z is at least

$$(1 - 2\varepsilon)(d(X, Y) - \varepsilon)(d(X, Z) - \varepsilon)(d(Y, Z) - \varepsilon)|X||Y||Z|$$

Proof. Let $X_R \subseteq X$ be the subset of vertices in X having at least $(d(X, Y) - \varepsilon)|Y|$ neighbours in Y and $(d(X, Z) - \varepsilon)|Z|$ neighbours in Z . By definition, every vertex in $X \setminus X_R$ fails in one of these two regards.

Let X_{NY} be the set of vertices failing the first condition. If $|X_{NY}| \geq \varepsilon|X|$, then $X_{NY} \subseteq X, Y \subseteq Y$ contradicts the ε -regularity of (X, Y) . The same is true for the similarly defined set X_{NZ} of vertices failing the second condition. As $X \setminus X_R = X_{NY} \cup X_{NZ}$, $|X_R| \geq (1 - 2\varepsilon)|X|$ (with the inequality arising from the fact that X_{NY}, X_{NZ} are not necessarily disjoint.)

Hence, let us count the number of triangles having elements of X_R as a vertex. Take any $x \in X_R$, and let $N(x, Y)$ be its neighbours in Y and $N(x, Z)$ be its neighbours in Z . Recalling that $\min(d(X, Y), d(X, Z), d(Y, Z)) \geq 2\varepsilon$, and combining this fact with our definition of X_R , we have that $|N(x, Y)| \geq \varepsilon|Y|, |N(x, Z)| \geq \varepsilon|Z|$. This means we can apply the ε -regularity of Y, Z to the two sets $N(x, Y), N(x, Z)$ to obtain at least $(d(Z, Y) - \varepsilon)|N(x, Y)||N(x, Z)| \geq (d(Z, Y) - \varepsilon)(d(X, Y) - \varepsilon)(d(X, Z) - \varepsilon)|Y||Z|$ triangles containing x . Counting over every choice of $x \in X_R$, and using the earlier fact that $|X_R| \geq (1 - 2\varepsilon)|X|$, we therefore obtain

$$(1 - 2\varepsilon)(d(X, Y) - \varepsilon)(d(X, Z) - \varepsilon)(d(Y, Z) - \varepsilon)|X||Y||Z|$$

triangles as desired. □

One consequence of these two lemmas is the following.

Lemma 4.10 (Triangle removal lemma) *For all $\varepsilon > 0$, there exists $\delta > 0$ such that any graph $G = (V, E)$ where $|V| > n$ containing less than δn^3 triangles can be made triangle-free by removing at most εn^2 edges.*

Proof. Consider a $\frac{\varepsilon}{4}$ -regular partition of the graph into m parts $\{V_1, \dots, V_m\}$. Now, remove all edges between V_i, V_j if any of the following hold:

- (i) (V_i, V_j) is not ε -regular;
- (ii) $d(V_i, V_j) \leq \frac{\varepsilon}{2}$;
- (iii) $\min(|V_i|, |V_j|) \leq \frac{\varepsilon n}{4m}$.

The first point leads to the removal of at most $\frac{\varepsilon n^2}{4}$ edges; the second, to at most $\frac{\varepsilon n^2}{2}$ edges and the third to at most $\frac{\varepsilon n^2}{4}$ edges, or a total removal of at most εn^2 edges. Now, for a given triple V_i, V_j, V_k , if there are still triangles having one vertex in each, then since we did not remove all edges between any two of these, the non-satisfaction of (a), (b) and (c) for removal actually implies we can apply the triangle counting lemma. So we have at least $(1 - \frac{\varepsilon}{2})(\frac{\varepsilon}{4})^3(\frac{\varepsilon}{4m})^3 n^3$ triangles, with the first term corresponding to the $(1 - 2\varepsilon)$ term from the counting lemma (recalling that we took a $\frac{\varepsilon}{4}$ -regular partition), the second term corresponding to the minimal density imposed once we removed edges and the third term corresponding to the minimal size once we removed edges. Double counting triangles over all unordered triples V_i, V_j, V_k (and dividing by 6 because of each triple appearing $3!$ times), it follows that

$$\delta = \left(1 - \frac{\varepsilon}{2}\right) \left(\frac{\varepsilon}{4}\right)^3 \left(\frac{\varepsilon}{4m}\right)$$

works. □

Finally, Theorem 4.4 follows as a consequence of this result. Indeed, let G_n be some graph on n vertices such that each edge is in exactly one triangle and $|G_n|$ is maximal. As every edge is in exactly one triangle, and we have less than $\frac{n^2}{2}$ edges, there are at most $\frac{n^2}{6} = o(n^3)$ triangles in G_n .

This means that, for any $\varepsilon > 0$, there is N such that for any graph $G_n = (V_n, E_n)$ on $n > N$ vertices such that each edge is in exactly one triangle and $|E_n|$ is maximal, one can apply the triangle removal lemma. Therefore, for $n > N$, removing εn^2 edges gives a triangle-free graph. Of course, since each edge is in a unique triangle, we must remove exactly one edge per triangle to achieve this, so $|E_n| \leq \varepsilon 3n^2$. Therefore

$$\lim_{n \rightarrow \infty} \frac{|E_n|}{n^2} \leq 3\varepsilon$$

and as this is true for all $\varepsilon > 0$ and the limit is positive, this limit is 0 as desired. This concludes the proof of Theorem 4.4. □

4.2 Bergelson's proof of Roth's theorem for abelian countable groups

In this section, we discuss Theorem 4.22 from Bergelson's 2000 paper[16], which is an ergodic proof of Roth's theorem for countable abelian groups.

4.2.1 Preliminaries

Firstly, we ask ourselves the following question: What would the statement of Roth's theorem be for a countable abelian group G ? Another formulation of the theorem for $\mathbb{N}$ is as follows: For any subset $E \subseteq \mathbb{N}$ with positive upper density, there exists $n \in \mathbb{N}$ such that $E \cap (E - n) \cap (E - 2n) \neq \emptyset$. But we can actually say something stronger than this, as we state now without proof.

Theorem 4.11 (Density Roth) *For any subset $E \subseteq \mathbb{N}$ with positive upper density, there exists $n \in \mathbb{N}$ such that*

$$\bar{d}(E \cap (E - n) \cap (E - 2n)) > 0.$$

We will now establish a variant of Theorem 4.11 for an arbitrary countable abelian group G . To do so, we will need some preliminaries; namely, re-establishing versions of many prior definitions for countable abelian groups.

Definition 4.12 (Measure-preserving dynamical system) *Given a group G , a probability space $(X, \mathcal{B}, \mu)$ and a G -action $\{T_g\}_{g \in G}$ acting on X such that, for all $g \in G$ and $A \in \mathcal{B}$,*

$$\mu(T_g(A)) = \mu(A),$$

we say the quadruple $(X, \mathcal{B}, \mu, \{T_g\}_{g \in G})$ is a measure-preserving dynamical system. For convenience, we denote the set $\{T_g\}_{g \in G}$ as T_G and refer to the quadruple as $(X, \mathcal{B}, \mu, T_G)$.

Definition 4.13 (Følner sequence) *Given a group G , a sequence of finite subsets $(F_i)_{i=1}^\infty$ of G is called a Følner sequence if, for all $g \in G$:*

$$\lim_{n \rightarrow \infty} \frac{|gF_n \triangle F_n|}{|F_n|} = 0. \quad (4.2)$$

Note that $\triangle$ denotes the symmetric difference of two sets.

A group is *amenable* if and only if it has at least one Følner sequence (as shown in Chapter 4 of Paterson's book[30]). It is well known that countable abelian groups are amenable, and therefore have at least one Følner sequence. Note that when the group is not abelian, the previous definition is often called a *left Følner sequence* to distinguish in which order we place F_n and g in (4.2).

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

Definition 4.14 (Upper density) Let G be a countable abelian group and $(F_i)_{i=1}^\infty$ a Følner sequence in G . For $E \subseteq G$, we define the upper density with respect to $(F_i)_{i=1}^\infty$ as

$$\bar{d}(E) = \limsup_{n \rightarrow \infty} \frac{|E \cap F_n|}{|F_n|}.$$

Definition 4.14 depends on the choice of Følner sequence $(F_i)_{i=1}^\infty$, but when this choice has been made clear we will omit reference to $(F_i)_{i=1}^\infty$ for convenience.

Proposition 4.15 Let G be a countable abelian group, $(F_i)_{i=1}^\infty$ a Følner sequence in G , and $E \subseteq G$. Then for all $g \in G$,

$$\bar{d}(E) = \bar{d}(gE).$$

Proof. We know that by (4.2),

$$\lim_{n \rightarrow \infty} \frac{|gF_n \cap F_n|}{|F_n|} = 1 \quad (4.3)$$

as $|F_n| = |gF_n \cap F_n| + \frac{|gF_n \triangle F_n|}{2}$. Then we have

$$\begin{aligned} \bar{d}(gE) &= \limsup_{n \rightarrow \infty} \frac{|gE \cap F_n|}{|F_n|} \geq \limsup_{n \rightarrow \infty} \frac{|gE \cap gF_n \cap F_n|}{|F_n|} \\ &\stackrel{(4.3)}{=} \limsup_{n \rightarrow \infty} \frac{|gE \cap gF_n|}{|F_n|} \\ &= \limsup_{n \rightarrow \infty} \frac{|E \cap F_n|}{|F_n|} = \bar{d}(E) \quad \square \end{aligned}$$

Recall now the definition of a syndetic set in $\mathbb{N}$ (Definition 2.3). It is not hard to see that it can be given the following equivalent form: a $S \subseteq \mathbb{N}$ is syndetic if there exists $k \in \mathbb{N}$ and a finite set $\{n_1, \dots, n_k\} \subseteq \mathbb{N}_0$ such that

$$(S - n_1) \cup \dots \cup (S - n_k) = \mathbb{N}.$$

In the vein of this formulation, we use the following definition for syndeticity in countable abelian groups:

Definition 4.16 (Syndetic set) Given a countable abelian group G , we say that $S \subseteq G$ is syndetic if there exists a finite set $\{g_1, \dots, g_k\} \subseteq G$ such that

$$(g_1 S) \cup \dots \cup (g_k S) = G. \quad (4.4)$$

Proposition 4.17 Given a countable abelian group G and a syndetic set $S \subseteq G$, we have $\bar{d}(S) > 0$.

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

Proof. Let G be a countable abelian group, $(F_i)_{i=1}^\infty$ a Følner sequence in G and $S \subseteq G$ a syndetic set. Let $\{g_1, \dots, g_k\}$ be taken in G so as to satisfy (4.4). We have

$$\begin{aligned} 1 = \bar{d}(G) &= \limsup_{n \rightarrow \infty} \frac{|G \cap F_n|}{|F_n|} \leq \sum_{i=1}^k \limsup_{n \rightarrow \infty} \frac{|g_i S \cap F_n|}{|F_n|} \\ &= \sum_{i=1}^k \bar{d}(g_i S) = k \bar{d}(S) \end{aligned}$$

whwhere we applied Proposition 4.15 to get the the last equality. So $\bar{d}(S) \geq \frac{1}{k} > 0$. $\square$

We can state a variant of Theorem 4.11 for countable abelian groups as follows.

Theorem 4.18 (Density Roth for countable abelian groups) *Let G be a countable abelian group and $(F_i)_{i=1}^\infty$ a Følner sequence in G . If $E \subseteq G$ such that $\bar{d}(E) > 0$, then there exists $g \in G$ such that*

$$\bar{d}(E \cap -gE \cap -2gE) = \bar{d}(E \cap -gE \cap gE) > 0.$$

We state now a version of the Furstenberg correspondence principle for countable amenable groups (we state it here for countable abelian groups), which is Theorem 6.4.17 in Bergelson's paper [16].

Theorem 4.19 (Furstenberg correspondence principle) *Let G be a countable abelian group and $(F_i)_{i=1}^\infty$ a Følner sequence in G . Let $E \subseteq G$ such that $\bar{d}(E) > 0$. Then there exists a measure-preserving system $(X, \mathcal{B}, \mu, T_G)$ and a set $A \in \mathcal{B}$ with $\mu(A) = \bar{d}(E)$ such that for any $k \in \mathbb{N}$ and $g_1, \dots, g_k \in G$ we have*

$$\bar{d}(E \cap g_1 E \cap \dots \cap g_k E) \geq \mu(A \cap T_{g_1} A \cap \dots \cap T_{g_k} A). \quad (4.5)$$

Finally, here is Bergelson's result:

Theorem 4.20 *Let G be a countable abelian group and $(X, \mathcal{B}, \mu, T_G)$ be a measure-preserving system. For any Følner sequence $(F_i)_{i=1}^\infty$ in G and any $A \in \mathcal{B}$ with $\mu(A) > 0$ we have*

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \mu(A \cap T_{-g} A \cap T_g A) > 0.$$

Proof. (Theorem 4.18 assuming Theorem 4.20) Let G be a countable abelian group, $(F_i)_{i=1}^\infty$ a Følner sequence in G and $E \subseteq G$ such that $\bar{d}(E) > 0$. By Theorem 4.19, there exists a measure-preserving system $(X, \mathcal{B}, \mu, T_G)$ and a

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

set $A \in \mathcal{B}$ with $\mu(A) > 0$ satisfying (4.5) for any $k \in \mathbb{N}$ and $g_1, \dots, g_k \in G$. By Theorem 4.20, we have

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \mu(A \cap T_{-g}A \cap T_g A) > 0$$

implying, via (4.5), that

$$\liminf_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \bar{d}(E \cap gE \cap -gE) > 0.$$

In particular, there exists $g \in G$ such that

$$\bar{d}(E \cap -gE \cap gE) > 0$$

as desired. $\square$

Before we proceed to the proof of Theorem 4.20, we will need even more results and definitions. We have the following:

Definition 4.21 (Ergodic measure-preserving system) *A measure-preserving system $(X, \mathcal{B}, \mu, T_G)$ is called ergodic if it has the following property: For any set $A \in \mathcal{B}$ satisfying,*

$$\forall g \in G, \mu(A \Delta T_g A) = 0$$

we have $\mu(A) \in \{0, 1\}$.

The next result is a version of van der Corput's difference theorem due to Bergelson, McCutcheon and Zhang[31].

Theorem 4.22 *Let H be a Hilbert space and let $\{v_g\}_{g \in G}$ be a bounded family of elements of H indexed by a countable abelian group G . Let $(F_i)_{i=1}^\infty$ be a Følner sequence in G . If*

$$\lim_{N \rightarrow \infty} \frac{1}{|F_N|^2} \sum_{h, k \in F_N} \left| \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \langle v_{g+h}, v_{g+k} \rangle \right| = 0,$$

then

$$\lim_{n \rightarrow \infty} \left\| \frac{1}{|F_n|} \sum_{g \in F_n} v_g \right\| = 0.$$

For the next theorem (Theorem 4.23), we introduce first two definitions. In what follows, $(X, \mathcal{B}, \mu, T_G)$ is a measure-preserving system, H is the Hilbert space $L^2(X, \mathcal{B}, \mu)$ and for $g \in G$, we define $U_g: H \rightarrow H$ as the unitary operator defined by $(U_g f)(x) = f(T_g x)$.

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

Given $U_g: g \in G$ a unitary action of a countable abelian group G on H , and $(F_i)_{i=1}^\infty$ a Følner sequence in G , we define

$$\begin{aligned} H_c &:= \overline{\{v \in H: \text{the orbit } \{U_g v, g \in G\} \text{ is precompact in the norm topology}\}} \\ &= \overline{\text{Span}\{v \in H: \text{there exist } \lambda_g \in \mathbf{C} \text{ with } U_g v = \lambda_g v, g \in G\}} \\ H_{wm} &:= \{v \in H: \forall v' \in H, \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} |\langle U_g v, v' \rangle| = 0\}. \end{aligned}$$

Note the spaces H_c and H_{wm} are U_g -invariant for $g \in G$.

Theorem 4.23 (Koopman-von Neumann) *The set H_c is a closed subspace of H and its orthogonal complement is H_{wm} .*

We will be using the following two equivalent definitions: firstly, g is in H_{wm} when, for any $g' \in H$, we have

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|^2} \sum_{h, k \in F_n} |\langle U_h g, U_k g' \rangle|^2 = 0. \quad (4.6)$$

Secondly, $f \in H_c$ when we have

$$\forall \varepsilon > 0, \quad \{g \in G: \|U_g f - f\| < \varepsilon\} \text{ is syndetic in } G. \quad (4.7)$$

Finally, we have the following version of the von Neumann ergodic theorem for countable abelian groups.

Theorem 4.24 (A version of von Neumann for countable abelian groups) *Let H be a Hilbert space and let $\{U_g\}_{g \in G}$ be an antirepresentation of a countable abelian group G as a group of unitary operators on H (i.e. $U_{g_1} U_{g_2} = U_{g_2 g_1}$ for all $g_1, g_2 \in G$). Let $H_{inv} = \{f \in H: U_g f = f, \forall g \in G\}$ and let P be the orthogonal projection on H_{inv} . For any Følner sequence $(F_i)_{i=1}^\infty$ in G we have*

$$\lim_{n \rightarrow \infty} \left\| \frac{1}{|F_n|} \sum_{g \in F_n} U_g f - P f \right\| = 0.$$

4.2.2 Bergelson's proof

Bergelson notes that Theorem 4.20 follows naturally from the special case where $(X, \mathcal{B}, \mu, T_G)$ is ergodic. Therefore, we now endeavour to prove Theorem 4.20 for ergodic measure-preserving systems, and the curious reader may refer to Bergelson's paper to see how to extend the proof to non-ergodic measure-preserving systems. We will consider only the special case where G is a countable abelian group where $2G = \{g + g: g \in G\} = G$; Bergelson notes that only minor changes are necessary for the case when $2G$ is a subgroup of finite index, and the case where $2G$ is a subgroup of infinite index is left as an exercise to the reader.

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

Proof. (Theorem 4.20, the ergodic case when $2G = G$.) Let G be a countable abelian group; $(X, \mathcal{B}, \mu, T_G)$ an ergodic measure-preserving system and $(F_i)_{i=1}^\infty$ a Følner sequence in G . Since $2G = G$, we also know that $(X, \mathcal{B}, \mu, T_{2G})$ is ergodic. Let $H = L^2(X, \mathcal{B}, \mu)$ and $f = 1_A$ (recalling that 1_S denotes the indicator function of a set S). We shall show first that the limit

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} f(T_{-g}x) f(T_gx)$$

exists in the norm of H . If this is true, we then have that the limit

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \mu(A \cap T_{-g}A \cap T_gA) = \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \int_X f(x) f(T_{-g}x) f(T_gx) d\mu$$

also exists and we will only have to show it is positive to conclude. Let U_g be the unitary action on H induced by T_g for any $g \in G$: $(U_g f)x := f(U_g x)$, $f \in H$. By Theorem 4.23, we have a splitting $H = H_c \oplus H_{wm}$. Let $1_A = f_c + f_{wm}$ by this splitting. We will show that f_c is a bounded function. 1_A is real-valued and the operators $U_g, g \in G$ map real-valued functions onto real-valued functions, so f_c and f_{wm} are also real-valued. We see that $f_c \geq 0$ - indeed, f_c minimizes the distance from H_c to f and the function $f_{c \geq 0} = \max(f_c, 0)$, which is a compact function, would also minimize this distance. Similarly, $f_{c \leq 1} = \min(f_c, 1)$ also minimizes this distance so $f_c \leq 1$. Now consider

$$\begin{aligned} \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f U_g f &= \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_c U_g f_c + \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_c U_g f_{wm} \\ &\quad + \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_{wm} U_g f_c + \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_{wm} U_g f_{wm} \end{aligned}$$

By Theorem 4.24, the limits of the first three expressions exist; we will illustrate why for one of the three and the same will hold true by an analogous reasoning for the other two. Consider

$$\frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_c U_g f_{wm}.$$

Since f_c is a (potentially infinite) linear combination of eigenfunctions - namely, functions φ satisfying for some $\lambda_g \in \mathbb{C}, g \in G$ the equation $U_g \varphi = \lambda_g \varphi, g \in G$, it is enough to prove that for each such φ the limit

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} \varphi U_g f_{wm}$$

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

exists in norm. Notably, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} \varphi U_g f_{wm} &= \varphi \cdot \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \lambda_{-g} U_g f_{wm} \\ &= \varphi \cdot \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \tilde{U}_g f_{wm} = \varphi \cdot P f_{wm} \end{aligned}$$

where P is the orthogonal projection on the space of invariant functions of the action $\tilde{U}_g := \lambda_{-g} U_g, g \in G$. It is easy to see that since $\tilde{U}_g \varphi$ implies $U_g \varphi = \lambda_g \varphi$, the projection $P f_{wm}$ belongs to H_c . But $f_{wm} \perp H_c$ and we get $P f_{wm} = 0$. It remains to show that

$$\frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_{wm} U_g f_{wm} = 0 \quad (4.8)$$

which we will use Theorem 4.22 and (4.6) for. Let $v_g := U_{-g} f_{wm} U_g f_{wm}$. We have, by the ergodicity of $\{U_{2g}\}_{g \in G}$,

$$\begin{aligned} \frac{1}{|F_n|} \sum_{g \in F_n} \langle v_{g+h} v_{g+k} \rangle &= \frac{1}{|F_n|} \sum_{g \in F_n} \int_X U_{-g-h} f_{wm} U_{g+h} f_{wm} U_{-g-k} f_{wm} U_{g+k} f_{wm} d\mu \\ &= \frac{1}{|F_n|} \sum_{g \in F_n} \int_X U_{-g} (U_{-h} f_{wm} U_{-k} f_{wm}) U_g (U_h f_{wm} U_k f_{wm}) d\mu \\ &= \frac{1}{|F_n|} \sum_{g \in F_n} \int_X (U_{-h} f_{wm} U_{-k} f_{wm}) U_{2g} (U_h f_{wm} U_k f_{wm}) d\mu \end{aligned}$$

whose limit is, as $n \rightarrow \infty$, $\rightarrow \int_X U_{-h} f_{wm} U_{-k} f_{wm} d\mu \int_X U_h f_{wm} U_k f_{wm} d\mu$

$$= \left(\int_X U_h f_{wm} U_k f_{wm} d\mu \right)^2 = |\langle U_h f_{wm}, U_k f_{wm} \rangle|^2.$$

Now we use the characterization of H_{wm} (and (4.6)):

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|^2} |\langle U_h f_{wm}, U_k f_{wm} \rangle|^2 = 0.$$

It follows that (4.8) holds, and that

$$\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f U_g f = \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} U_{-g} f_c U_g f_c.$$

4.2. Bergelson's proof of Roth's theorem for abelian countable groups

Recalling that H_c is closed under products and that $H_c \perp H_{wm}$, we now have

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \mu(A \cap T_{-g}A \cap T_gA) &= \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \int_X f(x) f(T_g x) f(T_{-g} x) d\mu \\
 &= \int_X f \cdot \left(\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} U_g f U_{-g} f \right) d\mu \\
 &= \int_X (f_c + f_{wm}) \left(\lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} U_g f_c U_{-g} f_c \right) d\mu \\
 &= \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \int_X f_c U_g f_c U_{-g} f_c d\mu.
 \end{aligned}$$

We already showed that $f_c(x)$ - and by extension $U_g f_c = f_c(T_g(x))$ - was a positive non-zero real-valued function. By (4.7) we therefore have that for all $\varepsilon > 0$, the set

$$S_\varepsilon := \{g \in G : \|U_g f_c - f_c\| \leq \varepsilon\}$$

is syndetic (and, by Proposition 4.17, has positive upper density). Note that $g \in S_\varepsilon \iff -g \in S_\varepsilon$, since U_g is a unitary operator. Take ε sufficiently close to 0 such that

$$\delta := \int_X f_c \max(f_c - \varepsilon, 0)^2 d\mu > 0$$

which is possible as f_c is not zero everywhere. Then

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n} \int_X f_c U_g f_c U_{-g} f_c d\mu &\geq \limsup_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n \cap S_\varepsilon} \int_X f_c U_g f_c U_{-g} f_c d\mu \\
 &\geq \limsup_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n \cap S_\varepsilon} \int_X f_c \max(f_c - \varepsilon, 0)^2 d\mu \\
 &\geq \limsup_{n \rightarrow \infty} \frac{1}{|F_n|} \sum_{g \in F_n \cap S_\varepsilon} \delta \\
 &= \limsup_{n \rightarrow \infty} \frac{|F_n \cap S_\varepsilon|}{|F_n|} \delta = \bar{d}(S_\varepsilon) \delta > 0. \quad \square
 \end{aligned}$$

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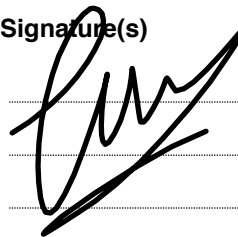
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